Heliyon 10 (2024) e36578

Contents lists available at ScienceDirect

Heliyon

journal homepage: www.cell.com/heliyon

Review article

A bibliometric analysis of IoT applications in logistics and supply chain management

Imen Zrelli [a,**], Abderahman Rejeb [b,*]

[a] University of Jeddah, Jeddah, Saudi Arabia
[b] Faculty of Business and Economics, Széchenyi István University, 9026, Győr, Hungary

ARTICLE INFO

Keywords:
Internet of things
Logistics
Supply chain management
Industry 4.0
Bibliometrics

ABSTRACT

The rapid integration of the Internet of Things (IoT) into logistics and supply chain management (SCM) marks a significant transformation towards enhanced efficiencies, security, and sustainability. Through a comprehensive bibliometric analysis of 2680 publications from the Scopus database, this study charts the evolution of IoT within logistics and SCM and reveals a shift from foundational explorations to mature implementations. The research unfolds a complex thematic structure, highlighting the revolutionary impacts of IoT and related technologies such as RFID, the synergy of Industry 4.0 with SCM through Artificial Intelligence (AI) and the Industrial Internet of Things (IIoT), the strategic role of blockchain for enhanced traceability and security, and the advent of novel communication and encryption technologies for secure data exchanges. Further, the analysis categorizes the scholarly discourse into critical areas including big data and IoT optimization in SCM, IoT-driven innovation in the food supply chain, applications of blockchain and smart contracts, digital transformation through Industry 4.0, security advancements with intelligent systems, and the exploration of advanced technologies for Industry 4.0 and 5.0. This review not only delineates the intellectual landscape of IoT applications in logistics and SCM but also identifies emerging research areas such as blockchain integration, 5G potential, and AI-driven optimizations, suggesting pathways for future research to broaden the understanding in this dynamically evolving field. It serves as an essential resource for academics and practitioners, providing insights into the transformative role of IoT in logistics and SCM and proposing directions for future technological and academic endeavors.

1. Introduction

The concept of the Internet of Things (IoT) is widely discussed and varies across different sources [1–3]. According to Atzori et al. [4], IoT represents a technology paradigm where various devices are interconnected and capable of being switched on and off from the internet, utilizing software and automation for intelligent applications. This interconnectivity is exemplified through mechanisms like RFID tags, which join networks to relay identification details [5,6]. Within the logistics and supply chain management (SCM) sphere, IoT is seen as a network of tangible objects with digital connections, designed to sense, monitor, and interact both within an organization and across its supply chain. This network fosters agility, transparency, tracking, and the exchange of information, thereby

* Corresponding author.
** Corresponding author.
E-mail addresses: Ibzrelli@uj.edu.sa (I. Zrelli), rejeb.abderahman@sze.hu (A. Rejeb).

https://doi.org/10.1016/j.heliyon.2024.e36578
Received 13 March 2024; Received in revised form 5 August 2024; Accepted 19 August 2024
Available online 20 August 2024
2405-8440/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Table 1
Examples of review studies on IoT research in logistics and SCM.

Study	Aims	Sample size (no. of papers)	Time span	Bibliometric techniques applied
[9]	To comprehensively analyze IoT research in SCM and logistics, identifying key trends and knowledge gaps to guide future research and inform practitioners of current applications and discussions.	807	2000–2020	Keyword co-occurrence network
[1]	To explore IoT's role and impact on SCM, reviewing literature across definitions, technologies, and SCM processes, while also categorizing studies and presenting a bibliometric analysis to highlight current focuses and future research areas in IoT implementation within SCM.	166	2008–2017	Co-citation network
[31]	To assess the impact of IoT on operations management by analyzing and classifying a vast array of scientific articles through bibliometric, life cycle, and text analyses, identifying key application clusters and influential contributors in the field.	1623	2000–2022	Reference co-citation network Article co-citation network Authors' collaboration network
[32]	To elucidate the role of IoT in SCM and logistics through a detailed literature review and bibliometric analysis, identifying technology enablers, key applications, and implementation challenges, while highlighting areas for future research to promote IoT implementation in various industries.	919	2000–2021	Co-authorship network Country collaboration network Keyword co-occurrence network
[37]	To address the knowledge gap in IoT-based SCM through a systematic literature review covering 2018–2022, providing a comprehensive analysis of applications, technologies, and challenges to guide future research and practice in the field.	130	2018–2022	None
[33]	To analyze the evolution of big data analytics and IoT within the context of SCM, identifying shifts in research focus and implementation over time.	689	2008–2017	Thematic network
[38]	To assess the impact of IoT technology on SCM performance through a systematic literature review and content analysis based on the Global Supply Chain Forum (GSCF) framework.	171	2000–2020	None
[34]	To explore the potential, challenges, and impact of IoT and other technologies like blockchain on food supply chain quality management through a literature review and bibliometric analysis.	144	2010–2019	Co-citation network
[35]	To explore the uncharted territory of IoT applications within the halal food supply chain (HFSC), aiming to bridge the knowledge gap by reviewing IoT research pertinent to HFSCs.	73	2008–2020	Keyword co-occurrence network Article co-citation network
[36]	To review the integration of Industry 4.0 technologies like IoT, AI, and cloud computing in warehousing and logistics, aiming to consolidate scattered research and chart a course for future inquiries.	64	2011–2021	None
Our study	To capture the rapid evolution of logistics and SCM through IoT advancements by conducting a comprehensive review and bibliometric analysis of previous studies on IoT applications	2680	2000–2023	Keyword co-occurrence network Co-citation network

enhancing the planning, control, and coordination of SCM activities [7–9]. IoT has emerged as a vital component in improving SCM operations [10]. The technology is envisioned as a global network that links objects and sensors, managed and optimized via wired or wireless connections, or a combination of both [4]. IoT's architecture is divided into three main elements: internet-oriented (middleware), things-oriented (devices, sensors), and semantic-oriented (data knowledge) [11]. The internet-oriented layer focuses on the technologies and protocols for networking physical objects and making them accessible online [12,13]. The things-oriented layer involves devices and smart objects like sensors, actuators, and RFID tags that connect to the internet [14], while the semantic-oriented layer addresses the challenges related to the vast amounts of data generated by these smart objects and their web-based resource integration [15].

With the advancement of wireless technology, IoT has seen a surge in popularity [1] and garnered significant interest within the logistics and SCM sector [8,16,17]. The impact of IoT has been profound in promoting industrial automation, facilitating the convergence of various networks such as industrial sensor networks, RFID networks for logistics, and systems for both plant control and enterprise information management [9]. Additionally, IoT has enabled companies to enhance their operational efficiency, streamline their processes, and maintain a competitive edge [18–20]. By leveraging IoT, businesses can optimize the flow of information [21], achieve significant efficiency improvements throughout the supply chain [22], and enhance communication and integration both within and between organizations [23,24]. For instance, Zara, a leading global fashion retailer, has leveraged IoT to achieve flexible planning, effective replenishment strategies, reduced lead times, and minimized product variations [25].

The importance of IoT within the industrial sector continues to grow, a trend highlighted by updated forecasts and analyses. Mordor Intelligence projects the industrial IoT market to reach USD 503.07 billion by 2029, marking a substantial increase with an impressive CAGR of 34.41 % from its 2024 estimate of USD 114.68 billion [26]. This significant growth reflects the enhanced utilization of data from connected devices, powered by advancements in big data and machine learning technologies. Additionally, Market Data Forecast anticipates the global industrial IoT market to expand at a CAGR of 11.91 %, expecting the market size to surge to USD

659.6 billion by 2029 from USD 375.7 billion in 2024 [27]. This expansion is indicative of the broader adoption of IoT devices and sensors in industrial settings for improved data collection and analysis.

Although IoT research has been extensively explored from the logistics and SCM perspective [28], there remains a significant gap in thoroughly examining the intellectual foundations and thematic progressions within this specific field. As a result, our investigation aims to bridge this gap through innovative bibliometric methods to dissect the influence of IoT on logistics and SCM, which is an approach not previously utilized. The current study embarks on an in-depth analysis of the IoT knowledge domain within logistics and SCM by scrutinizing co-citation patterns and keyword co-occurrence networks. We seek to offer a fresh perspective on the structural and network dynamics of IoT research in this area. The significance of our study lies in its potential to enhance the understanding of how IoT applications have evolved and influenced logistics and SCM. By leveraging bibliometric analysis, we can uncover new insights, highlight seminal works, and delineate emerging research trends and traditions. This comprehensive approach enables us to map the intellectual landscape of IoT in logistics and SCM, identifying key contributions and pivotal scholars who have shaped the field.

Our research is motivated by the insights of Portugal Ferreira [29], who underscores the necessity of continual literature review as fields mature and grow in complexity. Given that IoT technology rapidly advances, it is crucial to regularly reassess the accumulated scholarly work to identify significant contributions and propose new directions for future investigations. By delving into the network dynamics and structural nuances of IoT-related research in logistics and SCM, we aim to provide a comprehensive overview that not only chronicles past developments but also stimulates the conceptual evolution of this critical research domain. Furthermore, Wang et al. [30] argue for the importance of regular reviews to keep scholars informed of the rapid advancements in technology. In alignment with this perspective, our study addresses several key questions to further the understanding of IoT's impact on logistics and SCM. These include identifying core themes, major contributions, and future research directions that will benefit both academics and practitioners in the field. Overall, our study aims to fill a crucial gap in the literature by providing a targeted and comprehensive review

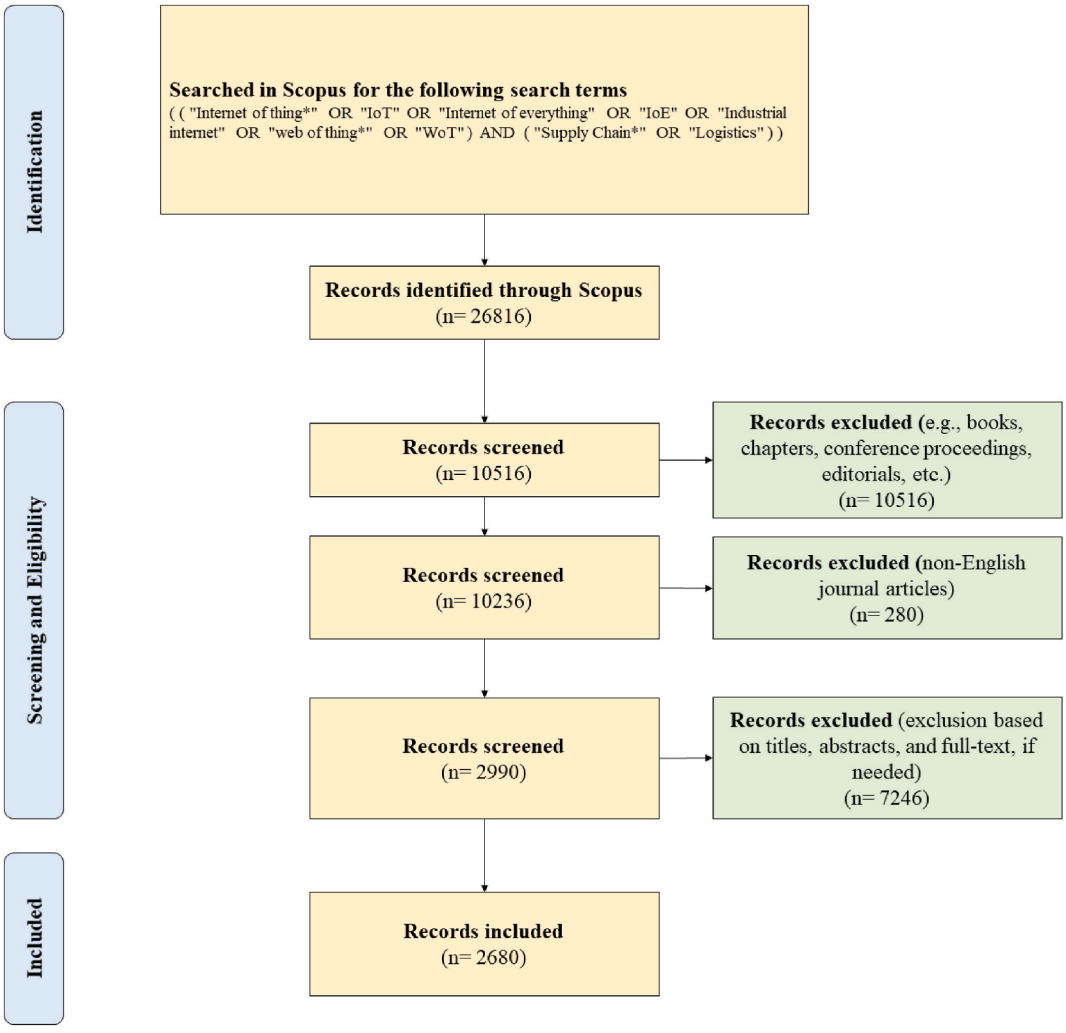


Fig. 1. Review process [45].

of IoT applications in logistics and SCM. Through innovative bibliometric methods, we seek to enhance the understanding of the field's intellectual and thematic progression, ultimately contributing to its ongoing development and practical application. Specifically, our study addresses several key research questions, including.

1) How has IoT research within logistics and SCM evolved from its inception?
2) Which journals, authors, institutions, and countries are leading in the advancement of IoT research in logistics and SCM?
3) What are the key themes and technological advancements in the application of IoT within logistics and SCM as identified through a keyword frequency analysis?
4) Which are the most influential studies on IoT applications in logistics and SCM?
5) What distinct themes emerge from the keyword co-occurrence network analysis of IoT applications in logistics and SCM?
6) What thematic clusters emerge from the co-citation network analysis of IoT applications in logistics and SCM?

2. Review of review studies on IoT applications in logistics and SCM

The integration of IoT into logistics and SCM represents a pivotal shift towards more efficient, transparent, and responsive operations. This literature review delves into the body of research exploring IoT applications within this domain, drawing insights from several studies to offer a comprehensive overview of how this technology is reshaping the field (Table 1). The work of Rejeb et al. [9] sets a foundational perspective through a bibliometric analysis spanning two decades, highlighting the significant attention IoT research has garnered within the SCM and logistics community. Their analysis reveals a concentration on RFID technology, Industry 4.0, reverse logistics, and broad applications across sectors such as food, retail, construction, and pharmaceuticals. This study not only underscores the diversity and depth of IoT's impact but also points to leading journals and authors shaping the discourse.

Building on this, Ben-Daya et al. [1] extend the exploration by focusing on the role of IoT in SCM, noting a particular emphasis on delivery processes and the prominence of the food and manufacturing supply chains. The authors call attention to the inclination of the existing literature towards conceptualizing IoT's impact, with a noted scarcity of analytical models and empirical studies, thereby marking critical areas for further investigation. Rezaee et al. [31] shift the lens towards operations management and identify key areas such as digitization, digital operations, monitoring systems, tracking, and smartification as central to IoT applications. The authors bring to light the expansive potential of IoT in enhancing operational efficiencies, yet they also acknowledge the nascent stage of research in this area, advocating for a deeper examination of IoT's operational implications. Katoch [32] utilizes VOSviewer software for a bibliometric analysis, offering a structured categorization of the literature on IoT within SCM and logistics. This approach not only highlights the technological enablers and diverse application areas but also signals the necessity for continued research on effective IoT implementation strategies. The study of Aryal et al. [33] of big data analytics and IoT uncovers evolving research trends, noting a shift in focus towards customer service, supply chain networks, and performance. This transition underscores the dynamic nature of SCM research in response to disruptive technologies and suggests fertile grounds for future studies.

Focusing on the food supply chain, Ben-Daya et al. [34] examine the potential and challenges of integrating IoT and blockchain in quality management. Their findings reveal a gap in literature regarding models and decision support systems that leverage IoT data for enhanced decision-making, indicating potential directions for advancing FSC management. In the context of the halal food supply chain, Rejeb et al. [35] highlight IoT's transformative benefits in ensuring product traceability, supply chain efficiency, and the authentication of halal status. However, the identification of challenges such as technological limitations and regulatory barriers outlines a comprehensive framework for future research and technological adoption. Lastly, Kumar et al. [36] review IoT in warehousing and logistics and point to the fragmented nature of existing research, with a notable concentration of studies in developed countries. The call of the authors for more theory-based research in IoT signifies the ongoing need to deepen and expand the scholarly understanding of IoT's capabilities and limitations within warehousing and logistics.

Our literature review synthesizes the findings from prior systematic reviews to highlight the transformative impact of IoT on logistics and SCM. Key areas of focus include the diverse applications of IoT technologies like RFID and Industry 4.0 across various sectors, the need for empirical research to complement conceptual studies, and the critical role of IoT in operational management improvements. Significant contributions by leading authors such as Rejeb et al. and Ben-Daya et al. emphasize the interconnected nature of this research, while specialized studies reveal gaps in decision support systems and the integration of blockchain with IoT. Future research should address these gaps, focusing on empirical validation and the development of comprehensive models to fully leverage IoT's potential in enhancing supply chain efficiency and transparency.

Building on this synthesis, our research stands apart by employing an unprecedentedly broad corpus of literature, previously unexplored, alongside innovative analytical techniques that combine keyword co-occurrence network and co-citation analysis. This novel approach allows for a more nuanced understanding of the field's dynamics, revealing critical areas in need of deeper investigation. As the SCM and logistics sector stands on the cusp of significant evolution driven by IoT, our study illuminates the path forward. It underscores the necessity for targeted research endeavors to fill the existing theoretical and empirical gaps and to unlock the full potential of IoT in enhancing SCM processes, especially in areas that have received less attention. By doing so, we aim to provide both academics and practitioners with a comprehensive roadmap to navigate the complexities of IoT adoption and leverage its capabilities to revolutionize supply chain operations.

3. Research methodology

The initial exploration began with the Scopus database using a query strategy outlined in Fig. 1. Scopus was chosen because it has a

wider reach than other databases like Web of Science, ProQuest, and IEEE Xplore [39,40]. Its advanced features make it easy to retrieve and collect academic references [41]. Scopus is also known for its comprehensive indexing from leading publishers such as Springer, Elsevier, Emerald Insight, ACM, Taylor and Francis, and IEEE, ensuring a high level of credibility and trustworthiness [42].

The search honed in on titles, abstracts, and keywords using a comprehensive set of terms related to IoT, logistics, and SCM, yielding a total of 7412 entries as of January 19, 2024. To ensure the inclusion of relevant literature, no restrictions were placed on specific subject areas, thereby encompassing the broadest spectrum of pertinent studies. To maintain academic rigor and uniformity among the selected works [43], we applied specific inclusion criteria: only English-language journal articles and reviews were considered, thereby excluding conference papers, chapters, books, and other document types [44]. This filtration step significantly reduced the pool to 3069 documents. Further refinement based on language criteria led to the consolidation of 2951 articles retained for in-depth examination.

The credibility and trustworthiness of the data were assured through a systematic manual screening process undertaken by a team of three researchers. Each researcher independently analyzed the content for relevance to IoT applications in logistics and SCM. To reach consensus on the findings, any discrepancies in the selection process were discussed and resolved through regular team meetings. Features of the articles included in the database encompassed publication year, authorship, journal title, keywords, abstract, citation count, and geographic distribution of the research.

We used both qualitative and quantitative methods to analyze the data. For the quantitative analysis, we applied bibliometric techniques like co-citation analysis, keyword co-occurrence networks, and citation analysis. We utilized software tools such as VOSviewer and BibExcel to facilitate these techniques. These methods helped us identify prominent themes, influential authors, and emerging trends within the field. For the qualitative analysis, we conducted a detailed content review to contextualize the bibliometric findings, ensuring a robust understanding of the research landscape. This meticulous selection process culminated in the exclusion of 4732 documents, pinpointing 2680 articles directly relevant to our research theme. The collation of these articles' bibliographic details was systematically organized in a CSV file. The procedural blueprint of our search and selection methodology is visually represented in Fig. 1. By combining rigorous manual screening with advanced bibliometric analysis tools, we ensured that our review process was both comprehensive and reliable. This approach provided a solid foundation for identifying key research areas, gaps, and future directions in the integration of IoT in logistics and SCM.

3.1. Review procedure

The essence of conducting a literature review lies in its ability to systematically and objectively collate, scrutinize, and interpret the existing scholarly work on a given topic, ensuring the process is transparent and can be duplicated with ease [46,46]. A thorough literature review, which integrates both explicit and nuanced connections within the research, emerges from a meticulous examination of diverse sources and methodologies. The decision to integrate bibliometric analysis into our study was driven by several

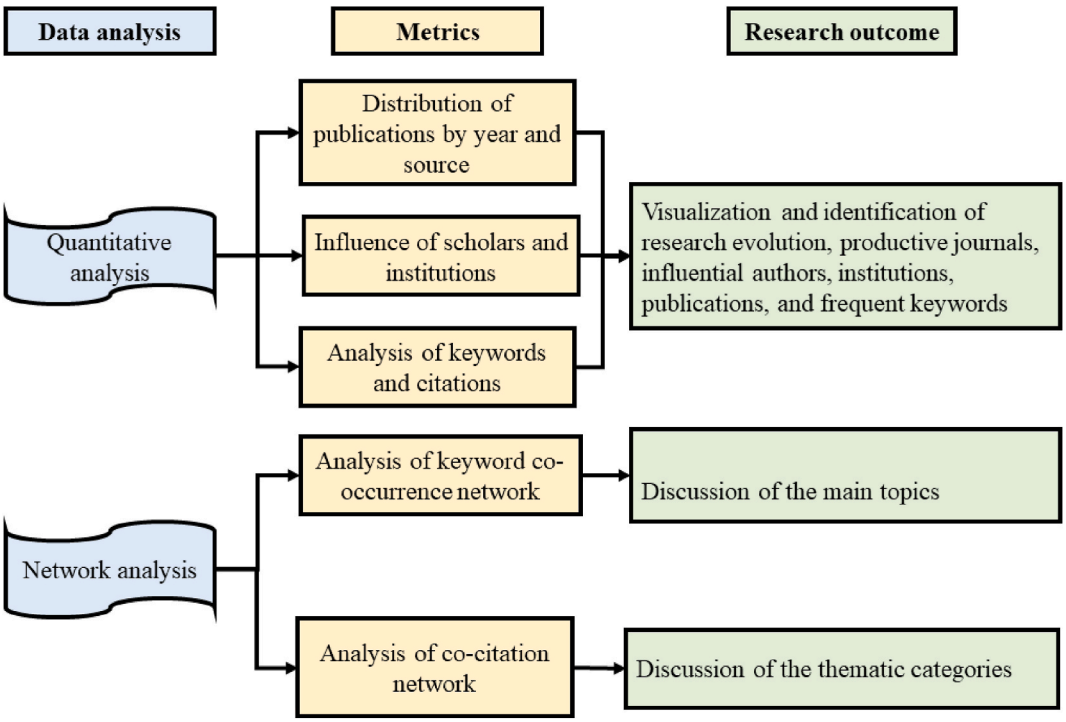


Fig. 2. Research approaches.

considerations. Firstly, bibliometrics offers a more reliable and expansive approach for textual exploration compared to traditional content analysis methods [47]. Secondly, it facilitates an in-depth exploration of the interconnections among publications, keywords, citations, and co-citations, thereby enriching the understanding of the research landscape [48,49]. Lastly, bibliometric techniques empower researchers to visually map out and interpret complex networks of research themes, presenting a clear and engaging overview of the field's intellectual terrain.

3.2. Quantitative analysis

Initially, our analysis began by charting the yearly distribution of the selected studies to trace the evolution of IoT research within logistics and SCM. Following the methodology outlined by Fahimnia et al. [50], we evaluated the impact and caliber of the research articles employing various quantitative indicators such as the authorship pattern, citation frequency per article, and the venues of publication. For data management and analysis, we utilized BibExcel, which is a versatile tool designed for the efficient handling of bibliographic data. BibExcel's compatibility with multiple scholarly databases (including Scopus and Web of Science) and its seamless integration with network analysis and visualization tools like VOSviewer, Ucinet, and Gephi [51], significantly enhanced our analytical capabilities.

Within the academic community, the number of citations a scholar receives is often viewed as a measure of their influence and contribution to their field. Accordingly, we organized the authors from our dataset based on their citation frequency. We utilized BibExcel to analyze the institutional affiliations of each author, identifying the leading academic institutions and the countries where they are based. To delve deeper into the research topics within IoT in logistics and SCM, a keyword analysis was performed to spotlight the most prevalent terms across the collected publications. Additionally, we examined citation patterns to gauge the impact and relevance of the research [52]. The extent to which a publication is cited serves as an indicator of its importance and its standing within the scholarly environment, reflecting its influence and reach [53].

3.3. Network analysis

After conducting a descriptive analysis of the selected articles, we ventured into identifying prevailing trends and connections within the body of work. Our methodology included employing a network analysis strategy, utilizing bibliometric data coupled with the visualization capabilities of Gephi to map out the intricate web of relationships inherent in IoT research within logistics and SCM. Initially, we constructed a keyword co-occurrence network, a technique that allows for the exploration of the frequency and context in which specific terms are mentioned together across various studies [54,55]. This method, as highlighted by Lee and Su [56], serves as a powerful tool in pinpointing pivotal research themes and tracing the evolution of scholarly discourse within a particular domain. In this network, the proximity of two keywords indicates a higher frequency of their concurrent appearance within the same documents, suggesting a strong relational tie. Utilizing VOSviewer for its superior integration with BibExcel data, we visualized the network, where the size of each node reflects the prevalence of a keyword, and the thickness of the links between nodes denotes the extent of keyword co-occurrence. This visual representation facilitates an intuitive understanding of the dominant themes and interconnections that characterize IoT research in logistics and SCM, offering insights into the current state and potential future directions of this rapidly evolving field.

A co-citation analysis has been a cornerstone method for assessing the semantic or conceptual similarities among publications that cite the same works [57]. This technique, alongside bibliographic coupling—which identifies papers sharing at least one reference or co-citation—serves as a foundation for mapping the intellectual structure of a field. The frequency of co-citations between two documents quantifies their relationship, with higher frequencies indicating stronger connections.

In this review, we adopted a co-citation analysis strategy, leveraging the citations within each selected article to delineate thematic clusters. Utilizing Gephi for visualization and BibExcel for data preparation, we meticulously crafted a co-citation network. The process

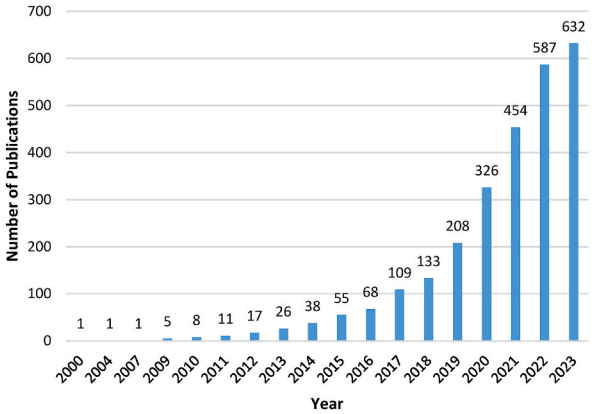


Fig. 3. Year-wise distribution of publications

Table 2
Most productive journals.

Journal	Number of Publications	%
Sensors	128	22 %
IEEE Access	118	21 %
Sustainability	84	15 %
IEEE Internet of Things Journal	58	10 %
Applied Sciences	39	7 %
Electronics	33	6 %
International Journal of Production Research	32	6 %
Computers and Industrial Engineering	30	5 %
Journal of Cleaner Production	25	4 %
Computational Intelligence and Neuroscience	22	4 %
	569	100 %

involved setting an appropriate co-citation frequency threshold to balance between overly dense and overly sparse clustering. Following recommendations from prior studies [42,58], we opted for a circle pack layout to achieve a clear and comprehensible network representation within Gephi. Each article within our dataset was represented as a node, with edges illustrating co-citations between pairs of articles. To refine the visualization, we manually adjusted various parameters such as node size, hierarchical relationships, and color coding, effectively curating a form of manual clustering and regularization of the data. Fig. 2 in our review depicts the methodologies employed to construct this nuanced map of research approaches, offering a visual exploration of the knowledge domains and interconnections within IoT research in logistics and SCM.

4. Quantitative analysis results

Understanding the evolution of IoT research within logistics and SCM is crucial for identifying trends, advancements, and gaps in the field. This context not only provides insight into how the field has developed over time but also highlights the transformative potential of IoT technologies in modernizing supply chains and logistics operations.

- *Research question 1: How has IoT research within logistics and SCM evolved from its inception?*

The progression of research on IoT applications in logistics and SCM over the years can be distinctly categorized into three phases of growth (Fig. 3). Initially, from 2000 to 2010, the field was in its infancy, marked by a sparse but foundational set of publications that laid the groundwork for future exploration. This period saw minimal activity, with research primarily focused on conceptual understandings and the potential of IoT to revolutionize logistics and SCM practices. As we moved into the period between 2011 and 2017, there was a noticeable shift towards early adoption and expansion. Publications grew from 11 to 109, reflecting an increased interest and practical exploration of IoT technologies in SCM, with studies showcasing pilot applications and the tangible benefits of IoT in enhancing supply chain efficiencies. The most transformative phase occurred from 2018 to 2023, where the field witnessed rapid growth and maturation, evidenced by a leap to 632 publications. This stage is characterized by a deepening and diversification of research, highlighting innovative IoT solutions for real-time tracking, inventory management, and optimization of supply chains [1,9]. Throughout these phases, the evolving landscape of IoT in SCM and logistics demonstrates a clear trajectory from theoretical exploration to essential strategic implementation and showcases the critical role of IoT in driving the future of supply chain innovations.

The identification of the leading journals, authors, institutions, and countries in IoT research within logistics and SCM is crucial for pinpointing key contributors and hubs of innovation. This insight is vital for recognizing areas of major advancements, guiding future research collaborations, and informing funding priorities.

Table 3
Most productive authors.

Author	Number of Publications	%
Huang GQ	28	23 %
Zhong RY	15	13 %
Luthra S	11	9 %
Tanwar S	10	8 %
Wu CH	10	8 %
Tsang YP	10	8 %
Gunasekaran A	9	8 %
Raut R	9	8 %
Qu T	9	8 %
Jagtap S	9	8 %
	120	100 %

Table 4
Most productive institutions.

Affiliation	Number of Publications	Location
The University of Hong Kong (HKU)	43	Hong Kong
The Hong Kong Polytechnic University (PolyU)	29	Hong Kong
King Abdulaziz University (KAU)	26	Saudi Arabia
King Saud University (KSU)	25	Saudi Arabia
Vellore Institute of Technology (VIT)	24	India
Ministry of Education of the People's Republic of China	21	China
Wuhan University	21	China
SRM Institute of Science & Technology	20	India
Fudan University	18	China
Shenzhen University (SZU)	17	China
National Institute of Industrial Engineering (NITIE)	17	India

Table 5
Most productive countries.

Country	Number of Publications	%
China	719	30 %
India	459	19 %
USA	303	12 %
UK	212	9 %
Saudi Arabia	146	6 %
South Korea	130	5 %
Australia	124	5 %
Italy	119	5 %
Germany	118	5 %
Pakistan	105	4 %
	2435	100 %

- *Research question 2: Which journals, authors, institutions, and countries are leading in the advancement of IoT research in logistics and SCM?*

Leading journals: The analysis of publications on IoT applications in logistics and SCM shows a concentration of research in key journals, with "Sensors," "IEEE Access," and "Sustainability" leading the field (Table 2). These journals, along with others like "IEEE Internet of Things Journal" and "Applied Sciences," account for a significant share of the research output, highlighting the technological focus of IoT studies in SCM. A similar pattern is observed in the healthcare sector, where "IEEE Access" and "IEEE Internet of Things Journal" are among the top publishers, indicating the interdisciplinary nature of IoT research [42]. This distribution underscores the pivotal role certain journals play in advancing IoT applications across both domains.

Leading authors: Table 3 highlights the contributions of leading authors in the field of IoT applications in logistics and SCM.

Table 6
Top 20 frequent keywords in the IoT and logistics and SCM.

Keyword	Number of Occurrence
IoT	1425
Blockchain	437
I4.0	293
SC	225
ML	191
SCM	191
RFID	162
Logistics	111
AI	109
IIoT	108
Big Data	107
Cloud Computing	87
Sustainability	86
Traceability	80
Smart Contracts	78
Security	77
CPS	76
FSC	61
Circular Economy	56
Deep Learning	56

Huang GQ emerges as the foremost contributor with 28 publications, representing 23 % of the analyzed works and illustrating a dominant presence in IoT research. Following closely, Zhong RY has produced 15 publications, accounting for 13 %, and Luthra S with 11 publications or 9 % of the total. Notably, Tanwar S, Wu CH, and Tsang YP each have contributed 10 publications, each constituting 8 % of the dataset. Additionally, Gunasekaran A, Raut R, Qu T, and Jagtap S have made significant impacts, each with 9 publications or 8 % of the research output. Together, these top authors are responsible for 4.4 % of the 2680 publications selected, showcasing a specialized focus and substantial expertise in leveraging IoT for enhancing logistics and SCM operations. This concentration of contributions from a select group of authors underscores the depth of investigation and development within the IoT SCM domain and reflects the evolving complexity and growing significance of IoT technologies in optimizing logistics and supply chain processes.

Leading academic institutions: Table 4 reveals significant insights into the landscape of IoT applications in logistics and SCM, focusing on the contributions of academic institutions and the geographical distribution of research productivity. The University of Hong Kong (HKU) and The Hong Kong Polytechnic University (PolyU) lead as the most productive institutions, with 43 and 29 publications respectively, underscoring Hong Kong's pivotal role in advancing IoT research in logistics and SCM. Additionally, King Abdulaziz University (KAU) and King Saud University (KSU) in Saudi Arabia, along with the Vellore Institute of Technology (VIT) in India, highlight the global nature of IoT research, with significant contributions emanating from Asia and the Middle East. Moreover, the presence of multiple institutions from China and India in the top ranks reflects the strong emphasis these countries place on IoT applications in logistics and SCM. The analysis also points to a broader trend of IoT research flourishing in developing countries, as evidenced by the list of most productive countries, where China and India lead with 719 and 459 publications, respectively. This indicates a robust interest and investment in IoT technologies as a means to enhance logistics and SCM efficiencies in these regions.

Leading countries: In terms of geographical distribution, the analysis underscores the dominance of Asian countries in IoT research, with China, India, and Saudi Arabia featuring prominently (Table 5). This trend suggests a strategic focus on IoT technologies in these countries, driven by the potential benefits of IoT in transforming logistics and SCM practices. The significant research output from these regions reflects a concerted effort to leverage IoT for economic development and competitive advantage in the global SCM arena. In conclusion, the analysis highlights the dynamic and global nature of IoT research, with leading contributions from institutions and countries that are actively exploring innovative IoT applications to address complex logistics and SCM challenges. The focus on developing countries emphasizes the critical role of IoT technologies in driving supply chain efficiencies, sustainability, and resilience, particularly in regions poised for rapid industrial growth and technological advancement.

Scholars are required to recognize the key themes and technological advancements in IoT applications within logistics and SCM to understand the current landscape and future directions of research. This contextual insight provides a foundation for identifying areas of significant impact and potential growth in the field.

- *Research question 3: What are the key themes and technological advancements in the application of IoT within logistics and SCM as identified through a keyword frequency analysis?*

Analyzing the top 20 frequent keywords from the literature on IoT applications in logistics and SCM (Table 6), "IoT" emerges as the most prevalent keyword with 1425 occurrences, underscoring the central role of IoT technology in modernizing supply chains and logistics operations [1,17]. The prominence of "Blockchain" with 437 occurrences reflects the growing interest in secure, transparent transaction frameworks within the supply chain and highlights the technology's potential to enhance traceability and trust between parties [2,59,60]. The term "I4.0" (Industry 4.0), noted 293 times, signifies the integration of IoT with other revolutionary technologies to drive the fourth industrial revolution, emphasizing smart, automated systems in logistics and SCM [61–63]. Keywords such as "SC" (Supply Chain) and "SCM" (Supply Chain Management), each appearing 191 times, along with "Logistics" (111 occurrences), underscore the research focus on the entire supply chain ecosystem, from management practices to logistical operations, emphasizing efficiency and optimization. The mention of "RFID" (162 times) points to the importance of real-time tracking technologies in enhancing inventory management and operational visibility [6,64]. The occurrence of "AI" (Artificial Intelligence) 109 times, alongside "ML" (Machine Learning) 191 times, indicates a significant focus on how intelligent algorithms can predict demand, optimize routes, and improve decision-making processes within logistics and SCM [65,66]. "IIoT" (Industrial Internet of Things) with 108 occurrences further highlights the application of IoT technologies in industrial settings, driving automation and connectivity in manufacturing and logistics.

Table 7
Top 10 most cited articles.

Citation	Title	Citations
[73]	The Internet of Things (IoT): Applications, investments, and challenges for enterprises	1843
[74]	Big Data in Smart Farming – A review	1475
[75]	Industry 4.0 technologies: Implementation patterns in manufacturing companies	1433
[76]	Industry 4.0 and the current status as well as future prospects on logistics	1123
[77]	Internet of Things - New security and privacy challenges	1032
[78]	Smart manufacturing	799
[79]	"Industrie 4.0" and smart manufacturing-a review of research issues and application examples	786
[1]	Internet of things and supply chain management: a literature review	720
[80]	Industry 4.0 and the circular economy: a proposed research agenda and original roadmap for sustainable operations	656
[2]	Can Blockchain Strengthen the Internet of Things?	626

"Big Data" (107 occurrences) and "Cloud Computing" (87 occurrences) reflect the critical role of data analytics and cloud infrastructure in managing vast amounts of data generated by IoT devices, enabling scalable, flexible logistics and SCM solutions. "Sustainability" (86 times) and "Circular Economy" (56 times) emphasize the growing concern for environmentally sustainable practices within SCM, driven by IoT's ability to monitor and reduce waste [67,68]. "Traceability" (80 times) and "Smart Contracts" (78 times) are indicative of the demand for transparency and automated agreements in supply chains, facilitated by technologies like blockchain [69, 70]. "Security" (77 times), a key concern, highlights the imperative to protect sensitive data within IoT-enabled logistics and SCM systems. Lastly, "CPS" (Cyber-Physical Systems) with 76 occurrences, and "FSC" (Food Supply Chain) with 61, indicate niche areas of IoT application, focusing on the integration of computational algorithms with physical processes and specific supply chain sectors like food, respectively [16,71,72]. This keyword analysis not only sheds light on the current state of IoT in SCM and logistics, highlighting the technology's multifaceted applications but also points to key areas of concern and opportunity, such as security, sustainability, and the integration of advanced technologies for more resilient and efficient supply chains.

It is imperative to ascertain which studies have driven the field of IoT applications in logistics and SCM forward, to fully understand the intellectual underpinnings and main themes of the research. This assessment underscores the seminal works that have and will continue to shape the direction of the field.

- *Research question 4: Which are the most influential studies on IoT applications in logistics and SCM?*

The analysis of the top 10 most cited articles within the context of IoT applications in logistics and SCM reveals critical insights into the domain's intellectual foundation and key thematic focuses (Table 7). The leading article by Lee and Lee [73] on the broad applications and challenges of IoT in enterprises has garnered the highest citations, emphasizing the significant interest in understanding IoT's implications for business operations. Following closely, Wolfert et al.'s [74] review on the role of big data in smart farming underscores the importance of IoT in enhancing agricultural productivity and sustainability. The prominence of Industry 4.0 is highlighted by Frank et al. [75] and Hofmann and Rüsch [76], indicating a strong linkage between IoT and the impact of the fourth industrial revolution on manufacturing and logistics. These works collectively suggest a trend towards digital transformation in industries, facilitated by IoT technologies. Security concerns associated with IoT deployment are addressed in the work of Weber [77],

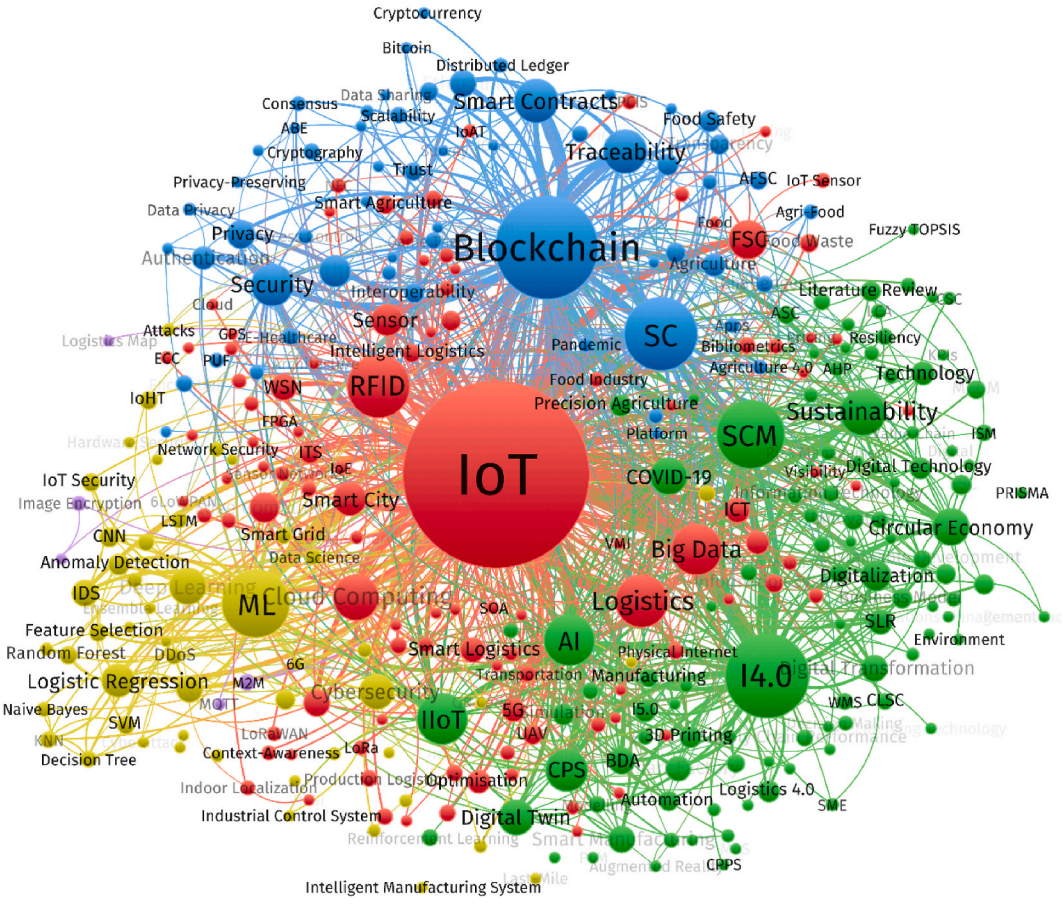


Fig. 4. Keyword co-occurrence network.

Table 8
Top 10 frequent keywords in each cluster.

1	2	3	4	5
IoT	I4.0	Blockchain	ML	M2M
RFID	SCM	SC	Deep Learning	Image Encryption
Logistics	AI	Traceability	Cybersecurity	Chaotic Map
Big Data	IIoT	Smart Contracts	Logistic Regression	Logistics Map
Cloud Computing	Sustainability	Security	ANN	MQTT
FSC	CPS	Healthcare	IDS	–
Smart City	Circular Economy	Privacy	Anomaly Detection	–
Sensor	COVID-19	Ethereum	CNN	–
Edge Computing	Digital Twin	Authentication	Data Mining	–
Fog Computing	Digitalization	Agriculture	Feature Selection	–

reflecting ongoing discussions about safeguarding data in an increasingly connected world. Similarly, the exploration of smart manufacturing by Kusiak [78] and the review of Industry 4.0 and smart manufacturing by Thoben et al. [79] illustrate the evolving nature of production systems through IoT integration.

5. Analysis of bibliometric networks

5.1. Keyword co-occurrence network

Through the use of keyword co-occurrence network analysis, distinct themes are identified, shedding light on the complex effects of IoT in logistics and SCM. This analytical approach is crucial for discovering knowledge gaps and key themes, thereby shaping the trajectory of future research and development in this field.

- *Research question 5: What distinct themes emerge from the keyword co-occurrence network analysis of IoT applications in logistics and SCM?*

Keyword co-occurrence analysis was utilized as an innovative clustering technique to unveil distinct themes within the realm of IoT applications in logistics and SCM, facilitating the examination of textual content in scholarly articles [81]. This approach allows for the exploration of various topics by analyzing the frequency of keyword pairings, shedding light on the multifaceted impacts of IoT within logistics and SCM. This bibliometric method identifies and aggregates common keywords found in the literature, enabling researchers to pinpoint knowledge gaps and dominant themes in this field of study [82]. The process began with extracting and refining keywords from selected articles, followed by employing VOSviewer for data analysis. We applied density-based spatial clustering with a full counting method to create the network, setting a threshold of at least five occurrences for each term to ensure a focused cluster analysis [83].

The analysis resulted in a network comprising five distinct clusters (Fig. 4), with the top 10 frequent keywords for each cluster presented in Table 8. In the network diagram, the size of a node represents the term's relative frequency, illustrating the prevalence of specific keywords within the scholarly discussion on IoT in logistics and SCM. Each node symbolizes a keyword, with its size indicating the frequency of co-occurrence with other terms. The proximity between two nodes, determined by their density, illustrates the closeness of the relationship between the keywords and provides insights into the interconnectedness of themes within the research landscape.

The examination of the top 10 frequent keywords within each cluster reveals significant insights into IoT applications in logistics and SCM. The clusters, as illustrated in the network, showcase distinct areas of focus in the field.

The first cluster underscores the critical role of technological advancements, particularly IoT, RFID, and logistics, in revolutionizing logistics and SCM. IoT emerges as a central theme, enabling real-time tracking and seamless data exchange across the supply chain, while RFID technology enhances inventory management through automated tracking [84–86]. The integration of big data and cloud computing is crucial for managing the vast data generated, facilitating advanced analytics for predictive insights into supply chain disruptions [21,87]. Additionally, the inclusion of terms like food supply chain (FSC), smart city, sensor, edge computing, and fog computing reflects the expanding scope of IoT applications, pointing towards more sustainable and efficient supply chain solutions. This cluster illustrates how digital technologies are transforming SCM into more intelligent, interconnected, and resilient systems, adept at addressing the dynamic demands of global markets and sustainability goals.

The second cluster centers on the intersection of Industry 4.0 (I4.0) and SCM, highlighting the transformative impact of Artificial Intelligence (AI), the Industrial Internet of Things (IIoT), and digitalization technologies. This focus reflects a shift towards smarter, more sustainable supply chains, where AI and IIoT drive efficiency and innovation [88,89]. The emphasis on sustainability and the circular economy points to a growing recognition of the need for eco-friendly practices within logistics and SCM [68]. The inclusion of COVID-19 indicates research interest in the pandemic's implications on supply chains [90–92], while digital twin technology represents the move towards virtual modeling for optimizing SCM operations [93,94]. Overall, this cluster reveals a trend towards integrating advanced technologies to create resilient, efficient, and environmentally conscious supply chains in the era of digital transformation. The third cluster delves into the synergy between blockchain technology and SCM, underscoring the pivotal role of

blockchain in enhancing traceability, security, and privacy within supply chains [69,95,96]. Keywords such as "Smart Contracts" and "Ethereum" suggest a focus on leveraging blockchain for automating contractual obligations and transactions, thereby ensuring integrity and transparency [97,98]. The prominence of "Traceability" and "Authentication" highlights the demand for verifiable and secure tracking of goods, from origin to consumer. Additionally, the inclusion of "Healthcare" and "Agriculture" points to the diverse application areas where blockchain's potential to safeguard sensitive data and ensure product authenticity is being explored [99,100]. This cluster illustrates a growing research interest in utilizing blockchain to address key challenges in supply chain management, aiming for more secure, transparent, and efficient supply networks.

The fourth cluster focuses on the advanced application of machine learning (ML) and AI techniques in enhancing cybersecurity and data analysis within various domains. It highlights key technologies such as deep learning, logistic regression, artificial neural networks (ANN), and convolutional neural networks (CNN), emphasizing their role in identifying patterns, predicting outcomes, and ensuring data security [65,101,102]. The mention of "Cybersecurity" alongside "IDS" (Intrusion Detection Systems) and "Anomaly Detection" points to the critical need for robust security measures capable of detecting and mitigating threats in digital environments [103,104]. "Data Mining" and "Feature Selection" are indicative of the process of extracting valuable insights from large datasets and selecting the most relevant features for model training, enhancing the efficiency and accuracy of predictive models. This cluster underscores the importance of ML and AI in driving innovations in cybersecurity and data analytics, showcasing their potential to transform data management and security strategies.

Finally, the fifth cluster explores cutting-edge communication and encryption technologies pivotal for secure and efficient data exchange in logistics and other interconnected systems. It emphasizes Machine to Machine (M2M) communication, which facilitates direct data transfer between devices, enhancing automation and efficiency in processes [105]. "Image Encryption" and "Chaotic Map" point towards advanced methods for securing digital images and data, ensuring privacy and protection against unauthorized access. The inclusion of "Logistics Map" suggests a focus on optimizing logistics operations through sophisticated mapping and routing algorithms. "MQTT" (Message Queuing Telemetry Transport), a lightweight messaging protocol, is highlighted for its role in enabling efficient, low-bandwidth communication between devices in IoT ecosystems [106,107]. This cluster signifies the importance of secure, reliable communication technologies in supporting the seamless operation of increasingly digital and automated systems.

5.2. *Co-citation network analysis*

The analysis of thematic clusters from co-citation networks provides detailed insights into the core research areas and concentrated themes within IoT applications in logistics and SCM, which assists in pinpointing key areas of interest and uncovering potential gaps in the research.

- *Research question 6: What thematic clusters emerge from the co-citation network analysis of IoT applications in logistics and SCM?*

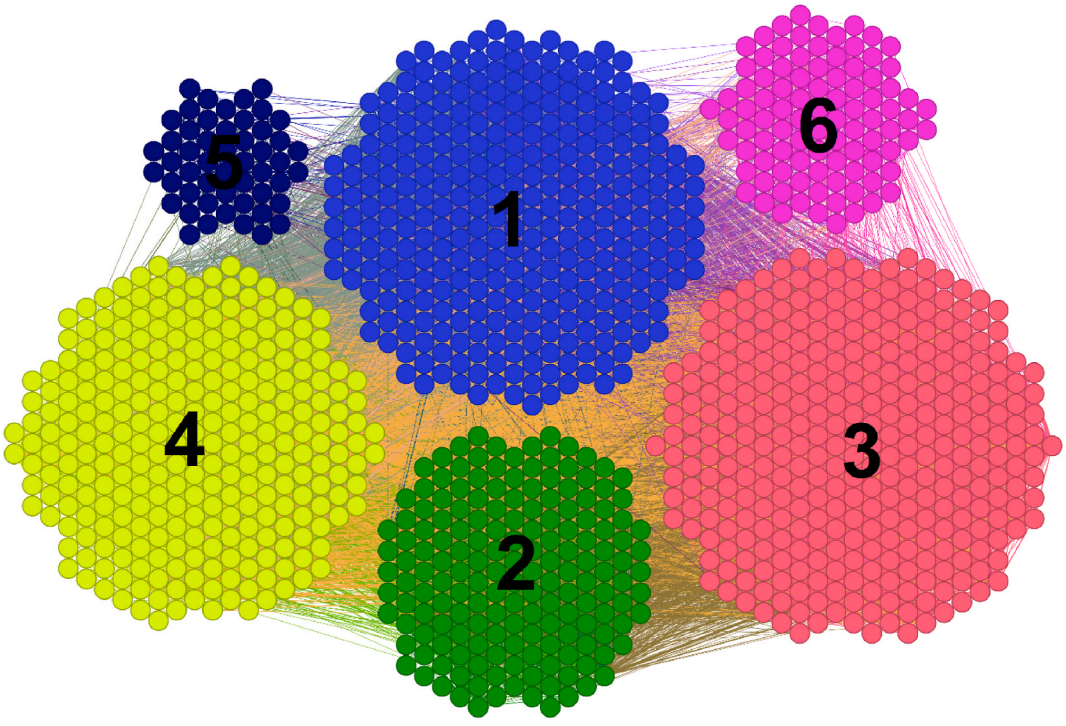


Fig. 5. Co-citation network.

In the context of IoT applications in logistics and SCM, the network analysis was conducted using Gephi's modularity tool, which employs the Louvain algorithm. This algorithm iteratively determines the most effective division of the network into clusters to maximize the modularity score. Modularity scores range from −1 to 1, reflecting the density of connections within clusters compared to those between clusters. For this study, the application of the algorithm revealed six distinct clusters with an overall network modularity of 0.388 (Fig. 5). This indicates a moderate level of interconnectivity among the clusters, each representing a unique facet of IoT research within logistics and SCM.

Clusters with a high level of co-citation suggest a concentrated research interest or theme. By examining the top ten articles from each cluster, we were able to identify and delineate the primary research areas within IoT applications in logistics and SCM. The key articles from each cluster, ranked by their influence, offer insights into the focal topics of current research. These areas of interest will be further explored in subsequent sections to provide a comprehensive overview of the state of IoT research in logistics and SCM.

5.2.1. Big data and IoT integration in supply chain optimization (cluster 1)

In the exploration of IoT applications within logistics and SCM, Cluster 1 delves into the synergistic integration of big data and IoT to optimize supply chains. This integration not only transforms logistics operations but also redefines manufacturing processes and delivery methodologies, showcasing a logical progression from theoretical frameworks to practical applications. The journey begins with the investigation of Zhong et al. [108] into the extension of the Physical Internet (PI) concept to manufacturing shop floors. By converting typical logistics resources into smart manufacturing objects (SMOs) through IoT and wireless technologies, the authors pioneer an RFID-enabled intelligent environment conducive to capturing vast quantities of RFID data. This foundational research underscores the importance of task weight in logistics decision-making, revealing insights into inventory management challenges within PI-enabled smart floors, thereby setting the stage for subsequent inquiries into data analytics within IoT-infused logistics environments.

Building on this premise, Hopkins and Hawking [109] conduct a case study of the application of big data analytics (BDA) and IoT in enhancing operational efficiencies of a large logistics firm. Focusing on driver safety, cost reduction, and environmental impact mitigation, the authors demonstrate the transformative potential of truck telematics and BDA in refining driving behaviors, optimizing routing, and forecasting maintenance needs. The seamless transition from Zhong et al.'s [108] theoretical model to Hopkins and Hawking's [109] real-world application exemplifies the logical progression from data capture to analytics-driven operational enhancements. Further extending this narrative, Zhong et al. [110] present a novel visualization approach for managing big data from RFID-enabled shop floors in cloud manufacturing settings. The RFID-Cuboid model introduced bridges the gap between complex data capture and user-friendly data interpretation, providing a tangible tool for operational decision-making.

Moreover, Zhu [111] researches cooperative logistical delivery scheduling using IoT and big data and further enriches the discourse by addressing the optimization of logistics resource allocation. By proposing a method that significantly improves the scheduling and utilization of resources, the author aligns with the overarching theme of operational efficiency, presenting a logical next step in the utilization of IoT and big data for supply chain enhancements. He et al. [112] explore IoT-enabled supply chain planning and coordination, adding a theoretical depth to the practical applications discussed earlier. According to the authors, the integration of IoT and big data brings several operational benefits and emphasizes the critical role of supply chain visibility in achieving these gains, thus reinforcing the interconnectedness of technological advancements and strategic SCM. Conclusively, Davis et al. [88] examine the interplay between industrial artificial intelligence, smart sensors, and big data-driven decision-making processes in real-time production logistics. By showcasing the application of structural equation modeling to understand these relationships, the authors underscore the practical implications of IoT and big data in enhancing supply chain and logistics operations. This logical progression from theoretical frameworks through practical applications and strategic implications highlights the transformative impact of IoT and big data integration in logistics and SCM, offering a coherent narrative that bridges the gap between innovative technologies and their real-world applications in supply chain optimization.

5.2.2. IoT-driven food supply chain innovation (cluster 2)

In the landscape of IoT-driven innovation within the food supply chain, the evolution of research demonstrates a compelling narrative that begins with foundational concepts and moves towards addressing complex challenges and creating value. This integrated approach underscores the transformative potential of IoT technologies, from enhancing operational efficiency and overcoming adoption barriers to ensuring food safety and generating new value propositions. The narrative of the second cluster embarks with the research of Verdouw et al. [113], who explore virtualization in food supply chains and emphasize how IoT enables dynamic management processes. This foundational study illustrates the capacity of IoT to support companies in managing perishable products, supply variations, and stringent safety requirements through real-time, remote monitoring and optimization. By applying this concept to a fish supply chain case study, Verdouw et al. [113] set the stage for future research into self-adaptive systems where smart objects autonomously operate, decide, and learn.

Building on the theme of technological adoption and implementation, Zhang et al. [114] introduce a conceptual model for an IoT-enabled perishable food supply chain. The authors demonstrate how real-time IoT data can enhance supply chain performance by focusing on the challenges specific to perishable food, such as cross-regional transportation, highlighting the practical application of IoT in improving freshness and reducing waste. Transitioning from operational enhancement to addressing barriers to technology adoption, Kamble et al. [115] delve into the challenges facing IoT integration in food retail supply chains, particularly within the Indian context. The authors identify significant barriers like the lack of government regulation and inadequate internet infrastructure, illuminating the complexities of IoT adoption, despite its potential to revolutionize operational efficiency, waste management, and energy consumption. In complement to the discussion on adoption barriers, Pang et al. [116] shift the focus toward the value-centric

design of IoT solutions. The authors advocate for creating income-centric values beyond traditional traceability and propose a business-technology joint design framework that aligns with new value propositions such as shelf life prediction and precision agriculture. The validation of this value-centric approach through field trials emphasizes the strategic importance of sensor portfolios and information fusion in realizing IoT's full potential.

Amidst operational and strategic considerations, Bouzembrak et al. [117] conduct a literature review and bibliometric analysis to explore IoT's role in food safety. By mapping out the landscape of IoT applications in food safety monitoring and quality control, this study not only highlights the technology's burgeoning impact but also identifies knowledge gaps, setting a course for future research endeavors. Further exploring the theme of food safety, Wang and Yue [118] introduce a pre-warning system based on data mining and IoT. This innovative approach to monitoring the entire supply chain in real-time exemplifies how IoT can proactively manage safety risks, offering decision support to ensure product quality and safety. Lastly, Li et al. [119] propose an IoT-based tracking and tracing platform to ensure the integrity of the food supply chain. By integrating QR code and RFID technologies within a service-oriented architecture, this platform offers an economical and effective solution for real-time management of prepackaged food supply chains, demonstrating the overarching potential of IoT in fostering a safer food consumption environment.

Together, these studies paint a comprehensive picture of how IoT is reshaping the food supply chain, from enhancing operational efficiencies and food safety to creating new value propositions and overcoming adoption barriers. The progression of research from virtualization and barrier identification to value creation and safety assurance underscores the transformative impact of IoT on the food supply chain landscape.

5.2.3. Blockchain and smart contracts in supply chain and IoT applications (cluster 3)

The third cluster delves into the intricate interplay between blockchain, smart contracts, and various application domains, revealing a multifaceted exploration of this technology's potential to revolutionize industries beyond its initial financial applications. This exploration begins with foundational insights into the principles and challenges of blockchain smart contracts, extends through innovative applications in logistics, agriculture, and healthcare, and culminates in addressing critical issues related to data integrity. The cluster includes the study of Lin et al. [120], which lays the groundwork by summarizing the operational principles, application status, and developmental challenges of blockchain smart contracts. This foundational study provides a panoramic view of the technological landscape, setting the stage for a deeper exploration of blockchain's capabilities and limitations. Lin et al.'s [120] analysis not only demystifies the technology but also forecasts its innovative applications and future development trends, offering a broad perspective that primes the reader for more specialized inquiries.

Building on this foundation, Hewa et al. [121] review the technical aspects and future research directions of blockchain-based smart contracts further refines our understanding. The authors elucidate the transformative potential of blockchain across various sectors. This detailed examination of technical nuances and potential advancements bridges the gap between theoretical constructs and practical applications, highlighting the versatility and adaptability of smart contracts. Transitioning from the theoretical to the practical, Hasan et al. [122] suggest a focused application of blockchain and smart contracts in enhancing efficiency within the global logistics and SCM. The development of a blockchain-based solution for tracking shipments via smart contracts illustrates the tangible benefits of integrating IoT with blockchain for real-time monitoring and management. This application not only exemplifies blockchain's utility in addressing logistical complexities but also showcases the operational and strategic advantages of smart contract implementation in global trade.

Further expanding the application spectrum, Pranto et al. [123] explore the integration of blockchain and smart contracts in smart agriculture. This innovative approach addresses traditional agricultural challenges by automating processes and establishing transparency and trust among all stakeholders. In the healthcare sector, Khatoon [124] present a blockchain-based smart contract system designed to streamline healthcare management. By reviewing existing applications and proposing new workflows for medical procedures and data management, the author highlights blockchain's capacity to secure data sharing and improve service delivery within the healthcare ecosystem. The feasibility study and cost analysis further underscore the practical implications and potential cost benefits of adopting blockchain in healthcare operations.

Ahmed et al. [97] study IoT data qualification for logistic chain traceability and introduce a critical consideration for blockchain applications: data integrity. By proposing a method to assess and ensure the quality of IoT data within a logistic traceability smart contract, Ahmed et al. [97] address a fundamental requirement for the successful deployment of blockchain technologies, emphasizing the importance of accurate, complete, and consistent data in achieving transparent and trusted systems. Concluding this exploration, Albizri and Appelbaum [125] tackle the Oracle Paradox, which is a significant challenge in the broader adoption of blockchain smart contracts. This research addresses the need for external validation of blockchain-recorded events and offers a solution to one of the most pressing issues facing smart contracts today. Overall, the studies in the third cluster articulate a comprehensive narrative of blockchain technology's evolution from a conceptual framework to a multifunctional tool capable of addressing specific challenges and revolutionizing practices in logistics, agriculture, healthcare, and beyond. This cluster stresses the potential of blockchain and smart contracts to transform industry standards and highlights the ongoing research and development efforts aimed at overcoming the technological and operational challenges inherent in these applications.

5.2.4. Industry 4.0: digital transformation and sustainability in logistics and SCM (cluster 4)

Cluster 4 extensively explores the pivotal role of Industry 4.0 in catalyzing digital transformation and enhancing sustainability within the sector. This group of studies underscores the integration of IoT and other Industry 4.0 technologies as a cornerstone for advancing logistics and SCM toward greater efficiency, transparency, and environmental responsibility. For example, Hofmann and Rüsch [76] initiate the discussion by identifying the transformative potential of Industry 4.0 in logistics management. The authors

outline the logistics-oriented applications of Industry 4.0 technologies and their implications, highlighting opportunities for decentralization, self-regulation, and efficiency improvements. Furthermore, Tjahjono et al. [126] delve into the specific impacts of Industry 4.0 on the supply chain and emphasize the enhanced collaboration and transparency achieved through cyber-physical systems and IoT. Their research underscores the necessity of adopting Industry 4.0 technologies for fostering a smart factory environment that significantly contributes to the efficiency and transparency of supply chain operations. Frank et al. [75] further expand on the adoption patterns of Industry 4.0 technologies in manufacturing firms, offering insights into the systemic implementation of these innovations. Their study reveals the central role of smart manufacturing within Industry 4.0 and the challenges companies face, particularly in leveraging big data and analytics, thus providing a nuanced understanding of technology adoption in enhancing logistics and SCM.

In a recent review, Manavalan and Jayakrishna [127] focus on the integration of IoT within sustainable supply chains to meet Industry 4.0 requirements and propose a comprehensive framework for assessing readiness for the fourth industrial revolution. This research highlights the critical importance of sustainability and technology in transforming supply chains to be more responsive and adaptable to market dynamics and environmental concerns. Lopes de Sousa Jabbour [80] advocate for the synergistic relationship between Industry 4.0 and the circular economy and propose a roadmap for employing advanced digital manufacturing technologies to facilitate sustainable operations. The authors illustrate how Industry 4.0 can underpin circular economy strategies and highlight the role of digital innovations in achieving resource circularity within supply chains. Similarly, Barreto et al. [128] introduce the concept of logistics 4.0 and address the technological shifts necessary for organizations to excel in a digitally transformed landscape. The authors highlight the critical requirements for achieving efficiency and operational excellence in the context of Logistics 4.0.

Winkelhaus and Grosse [129] unify the diverse research on logistics 4.0 into a coherent framework, defining the term and examining the technological innovations driving changes in logistics tasks. Finally, Fatorachian and Kazemi [130] investigate the impact of Industry 4.0 on supply chain performance and conceptualize an operational framework to assess the performance enhancements brought by Industry 4.0-enabling technologies. Their research emphasize the significant improvements in SCM processes, including procurement, production, inventory management, and retailing, facilitated by digital integration and automation. To conclude, the research focus of the fourth cluster articulates a compelling vision of digital transformation and sustainability in logistics and SCM, driven by the adoption of Industry 4.0 technologies. The cluster offers valuable insights into the challenges, opportunities, and future directions for organizations striving to navigate the complexities of a digitally transformed and environmentally conscious global market.

5.2.5. Intelligent systems and security in IoT for industry 4.0 (cluster 5)

The fifth cluster discusses the realm of emerging challenges and solutions in IoT security, focusing on the innovative approaches and strategies developed to fortify IoT networks against a growing spectrum of cyber threats. This cluster highlights the multifaceted nature of IoT security and explores both the challenges posed by the expansion of IoT devices and networks, and the cutting-edge solutions proposed to address these vulnerabilities. The studies within this cluster collectively underscore the critical importance of advancing IoT security mechanisms to safeguard interconnected devices and the data they process and store. For example, Sarker et al. [131] introduce the IntruDTree model, which is a novel machine learning-based cybersecurity intrusion detection system that emphasizes the prioritization of security features according to their importance. This approach not only enhances the prediction accuracy for detecting unauthorized access or anomalies but also significantly reduces computational complexity by focusing on the most relevant features.

Building on this, Abbas et al. [132] enrich the discourse by combining multiple machine learning algorithms (logistic regression, naive Bayes, and decision tree) through a voting classifier mechanism. The authors exemplify the strength of ensemble methods in improving detection accuracy, showcasing a strategic blend of different machine learning techniques to create a robust system capable of identifying malicious activities with greater precision. In a similar vein, Huma et al. [133] venture into the domain of deep learning with the introduction of a hybrid deep random neural network (HDRaNN) for cyberattack detection in the industrial IoT. Huma et al.'s [133] research is particularly noteworthy for its application of a deep learning framework that combines the strengths of random neural networks and multilayer perceptrons, achieving remarkable accuracy in classifying diverse types of cyberattacks.

The collective insights from these studies illuminate the evolving landscape of IoT security, characterized by a continuous search for more efficient, accurate, and computationally feasible solutions. The emphasis on machine learning and deep learning across these works reflects a broader trend towards leveraging AI to meet the challenges of securing increasingly complex and vast IoT networks. This cluster, therefore, contributes significantly to the broader discourse on IoT security by proposing innovative models that prioritize feature importance, harness the power of ensemble learning, and explore the frontiers of deep learning in detecting and mitigating cyber threats.

5.2.6. Advanced technologies in cyber-physical supply chains for industry 4.0 and industry 5.0 (cluster 6)

The exploration of advanced technologies in cyber-physical supply chains for Industry 4.0 and 5.0 begins with an understanding of the gradual evolution from Industry 4.0's emphasis on automation and digitalization to Industry 5.0's focus on human-machine collaboration. In this regard, Maddikunta et al. [134] introduce Industry 5.0 by highlighting the synergy between human creativity and machine efficiency, aiming for resource-efficient and user-preferred manufacturing solutions. This transition underscores a significant shift towards systems that not only prioritize efficiency but also customization and sustainability, thereby reflecting a broader trend in logistics and SCM towards more adaptive and responsive operations.

Building on this foundation, studies by Kamble et al. [135] and Andronie et al. [136] dive deeper into the technological innovations driving this evolution. The former study on digital twins showcases how this technology fosters sustainable, intelligent manufacturing systems by enabling real-time information flow throughout the product life cycle. Similarly, the latter study emphasizes the

importance of sustainable, smart, and sensing technologies, including AI and IoT, in enhancing the operational performance of cyber-physical manufacturing systems. These technologies contribute to creating automated, robust, and flexible production networks that can significantly improve SCM's efficiency and responsiveness.

As we further explore the applications of these technologies, Moshood et al. [137] illustrate the potential of digital twins to enhance supply chain visibility and offer predictive insights that can revolutionize asset management and operational planning. This is particularly relevant in the context of IoT applications in logistics and SCM, where real-time data and connectivity are crucial for optimizing logistics operations and ensuring the seamless flow of goods across global supply chains. However, the transition to Industry 5.0 and the integration of these advanced technologies into SCM are not without challenges. Alvarez-Aros and Bernal-Torres [138] highlight the need for developing skills, training, and digitalization efforts to fully leverage the benefits of IoT, AI, and digital twins in supply chains. Moreover, Verma et al. [139] address the critical issue of data integrity and security, emphasizing the role of blockchain as a key enabler for trusted data exchange in an increasingly interconnected and autonomous industrial landscape. In sum, the progression from Industry 4.0 to 5.0, marked by the integration of IoT, digital twins, AI, and blockchain technologies, offers promising avenues for transforming logistics and SCM. These technologies not only facilitate the move towards more sustainable, efficient, and customer-centric operations but also highlight the ongoing need to address the challenges of integration, data security, and workforce development to realize the full potential of Industry 5.0 in logistics and SCM.

6. Concluding remarks, research implications and future directions, limitations

6.1. Summary of findings

The integration of IoT is revolutionizing logistics and SCM, offering unparalleled opportunities to enhance operational efficiency, transparency, and customer satisfaction. IoT devices play a critical role in transforming logistics services by enabling real-time tracking of goods, predictive maintenance of transportation vehicles, and automated inventory management. This shift towards IoT-enabled logistics allows companies to optimize their delivery routes, reduce downtime, and ensure the timely delivery of goods, thereby significantly improving customer satisfaction and engagement. In SCM, IoT applications facilitate closer monitoring and management of the supply chain and allow for dynamic adjustments based on real-time data. For instance, IoT sensors can monitor the condition of goods in transit and ensure temperature-sensitive products are maintained within safe parameters, which is crucial in food and pharmaceutical logistics. This capability not only reduces the risk of spoilage and loss but also enhances the safety and reliability of the supply chain. Furthermore, the use of IoT in logistics and SCM has the potential to substantially decrease operational costs. By leveraging data from IoT devices, companies can predict potential disruptions in the supply chain, enabling proactive measures to mitigate risks and avoid costly delays. IoT-driven analytics can also identify inefficiencies in the supply chain, providing insights for optimization and cost reduction. Moreover, IoT enhances the personalization of logistics services. Through the continuous collection and analysis of data, logistics providers can offer customized delivery options, adapt services to meet individual customer preferences, and improve the overall customer experience. With the support of IoT, this level of personalization strengthens the relationship between logistics providers and their customers, fostering loyalty and trust. Additionally, the deployment of IoT technologies in logistics and SCM supports sustainability goals by optimizing routes to reduce fuel consumption and carbon emissions and by enabling smarter warehousing solutions that minimize energy use. As the logistics and SCM sectors continue to embrace IoT, the potential for innovation and efficiency gains appears boundless and promises a future where supply chains are more resilient, responsive, and aligned with the evolving needs of businesses and consumers.

Due to the rise of the literature on IoT and SCM and a paucity of systematic and thorough analyses of this topic, it is imperative to map the present status of existing knowledge. Therefore, this research was driven by the need to chart the development of this area of study and shed light on its underlying intellectual structure by mapping and visualizing the scope and structure of research at the confluence of IoT and logistics and SCM. Using bibliometrics, we studied 2680 documents drawn from the Scopus database over nearly a twelve-year period. By applying multiple bibliometric techniques, we were able to achieve unbiased findings. As Linnenluecke et al. [140] point out, this is contrary to subjective and biased techniques where "an arbitrary selection of evidence is often not fully representative of the state of the existing knowledge, and the selection of some studies over others ultimately leads to what is known in statistic analysis as a sample selection bias."

Given the increasing interest in IoT applications within logistics and SCM and the observed gap in comprehensive and systematic analyses in this domain, it becomes essential to delineate the current knowledge landscape. This study is propelled by the necessity to trace the evolution and intellectual foundation of IoT integration in logistics and SCM, aiming to illuminate the breadth and depth of research in this intersection. Leveraging bibliometric analysis, we scrutinized 2680 documents sourced from the Scopus database. The application of various bibliometric methodologies facilitated the attainment of impartial results, contrasting with the subjective nature of traditional review methods that may inadvertently introduce sample selection bias [140]. This bias occurs when the evidence is arbitrarily selected, not fully reflecting the extant knowledge and preferentially including some studies over others, thereby skewing the analytical outcome. Through our bibliometric approach, we sought to map and visualize the scope and structure of IoT in logistics and SCM research, offering a panoramic view that captures the convergence of technological advancements and supply chain dynamics. This endeavor aims not only to chart the progression of this research domain but also to uncover its underlying intellectual structure, providing a clear, unbiased representation of existing knowledge and identifying fertile areas for future inquiry.

To address the gaps and limitations inherent in conventional methodological approaches, which frequently fail to capture the intricate web of relationships and dynamics among researchers, publications, and academic journals within the logistics and SCM sectors, this study employs a comprehensive landscape of IoT research tailored to logistics and SCM, utilizing principles and techniques

from systematic network analysis. According to Chen and Leydesdorff [141], understanding the connections between these elements is crucial for a deep grasp of the conceptual foundations and historical evolution of knowledge within a specific scientific domain. To our knowledge, this investigation represents the inaugural endeavor to conduct an exhaustive bibliometric analysis focused on IoT applications within logistics and SCM, analyzing the entirety of scholarly works on the subject indexed in Scopus from 2011 onwards. The analysis of such a vast corpus of literature facilitates the identification of leading researchers, pivotal journals, and the principal trends and themes emerging within this cross-disciplinary research area, thereby providing a holistic view of the state of the art in IoT applications in logistics and SCM.

Our investigation into IoT applications in logistics and SCM reveals a dynamic field experiencing rapid growth, particularly notable since 2017. This upward trajectory in research output is expected to continue, reflecting the expanding role of IoT in these areas. Our analysis, focusing on publication outlets, shows a significant concentration of articles within computer science and technology journals, notably IEEE Access, IEEE Internet of Things Journal, and Sensors. These publications are instrumental in disseminating pioneering IoT research, thereby solidifying its importance in logistics and SCM innovations. In terms of contributions to the field, Huang GQ and Zhong RY stand out as the most prolific authors, indicating their substantial influence on the development of IoT applications in logistics and SCM. The University of Hong Kong emerges as a leading contributor to advancing IoT research. This suggests a global recognition of the importance of IoT technologies in transforming logistics and SCM practices. Geographically, the bulk of research originates from Asian countries, with China and India leading the charge. This not only underscores the technological ambition of these nations but also their strategic investment in IoT to drive efficiencies and innovations in logistics and SCM.

Our content analysis further identifies five distinct research clusters based on keyword co-occurrence, and seven clusters derived from co-citation networks. These clusters offer a nuanced understanding of the IoT's thematic and intellectual landscape within logistics and SCM, highlighting the diverse applications and theoretical advancements driving the field forward. This multifaceted view underlines the critical role of IoT in shaping the future of logistics and supply chain operations, marking it as a key area of focus for researchers and practitioners alike.

6.2. Research implications and future directions

In an effort to provide academics with a clear understanding of the advancements in IoT applications within logistics and SCM, a comprehensive analysis of literature was conducted focusing on the occurrence of specific keywords. Utilizing a bibliometric approach, this study unveiled the vast potential IoT holds in revolutionizing logistics and SCM towards more customized, intelligent, and efficient operations. Recent scholarly focus has shifted towards leveraging state-of-the-art technologies like AI, blockchain, digital twins, and machine learning, integrating these with IoT to enhance supply chain visibility, inventory management, and operational efficiency. This review meticulously selects significant publications, aiding scholars in deepening their comprehension of IoT's impact on logistics and SCM. The identification of seminal works and their authors not only sets the direction for future research but also encourages collaborative efforts among research communities to elevate the quality and innovation of their studies. Additionally, the keyword co-occurrence network analysis reveals the predominant themes and areas of interest within IoT research in logistics and SCM, similarly shedding light on the foundational and emergent topics driving the field forward.

To facilitate a deeper understanding of IoT's transformative impact on logistics and SCM, this study delves into the integration of IoT within these fields, particularly its role in advancing towards smart logistics and SCM systems. Future research is essential to dissect how IoT technologies are currently woven into logistics and SCM operations and their architectural implications. There is a pressing need to theorize on the potential of IoT to democratize preventive logistics services, making them more accessible, and to evolve secondary and tertiary logistics services into a seamless, proactive, and integrated framework. Surprisingly, discussions on the potential hurdles of IoT in logistics and SCM, including barriers to large-scale adoption from the perspectives of industry stakeholders, security, privacy concerns, trust and acceptance, interoperability, as well as data storage, ownership, and governance issues, are noticeably absent from existing literature. To successfully embed IoT into contemporary logistics and SCM frameworks, comprehensive policy support, cybersecurity standards, meticulous strategic planning, and explicit operational guidelines within logistics organizations are imperative. These measures will ensure the seamless incorporation of IoT technologies, paving the way for more intelligent, efficient, and sustainable logistics and supply chain systems.

Similarly, blockchain technology emerges as a novel solution capable of bolstering IoT deployment in logistics and SCM by ensuring transparency and security of data, facilitating product traceability, and enhancing the efficiency of data sharing. There is a pressing need for future research to test and validate the integration of blockchain and IoT in practical settings, offering tangible security assurances. The development of digital signatures resistant to quantum computing advances is crucial for mitigating cyber threats and safeguarding IoT-based logistics systems. Furthermore, the adoption of blockchain and IoT technologies in logistics and SCM could potentially lead to cost reductions, with the innovative concept of reimbursing or rewarding stakeholders in tokens, necessitating a foundational shift towards tokenization. Future investigations must also address how logistics organizations can overhaul their internal processes, enhance blockchain throughput, and devise effective consensus protocols tailored to the unique demands of blockchain and IoT solutions in logistics and SCM. Given the complex nature of logistics networks and the necessity for integrated sensor data across body area networks, the collaborative and connective fabric of IoT, alongside the development of data-centric consensus algorithms, becomes imperative. The keyword co-occurrence network underscores the significance of AI, digital twins, data analytics, and advanced computing technologies in augmenting IoT-based logistics and SCM. Consequently, forthcoming studies should delve into the convergence of IoT and AI to develop sophisticated models for predictive logistics management. The potential of 5G technology in IoT-based logistics warrants exploration from multiple angles, including system scheduling, machine learning applications, handover strategies, routing, and clustering algorithms. Moreover, future research should aim to create IoT solutions that

guarantee the integrity and accuracy of supply chain data. This is crucial for ensuring secure and straightforward interactions between smart logistics devices and cloud-based databases. Lastly, our co-citation network analysis highlights critical future research domains, such as authentication schemes for enhancing IoT security in logistics and paving the way to an integrated and technologically advanced future for logistics and SCM.

6.3. Research limitations

This review study acknowledges several limitations that must be considered for future research. Predominantly, the reliance on a single database, Scopus, for data acquisition presents a potential constraint on the comprehensiveness of the findings. Future studies are encouraged to incorporate data from additional databases (e.g., Web of Science) to enhance the robustness and validate the results of our analysis. Furthermore, our selection criteria were restricted to articles published in English, which may narrow the scope of the study and overlook significant contributions in other languages. To address this, future investigations should consider including publications in multiple languages to ensure a broader coverage and applicability of the study's insights. Another limitation arises from the possibility that pertinent literature may have been missed due to the specific search terms employed in this study. To mitigate this, it is advisable for future research to expand the range of search terms, incorporating additional keywords relevant to IoT applications in logistics and SCM. This approach will help capture a wider array of relevant studies, thereby enriching the understanding and analysis of IoT's impact on logistics and SCM. Moreover, this study suggests the adoption of alternative bibliometric methodologies, such as main path analysis and topic modeling, for subsequent review studies. These methods could offer new perspectives and deeper insights into the evolving trends, challenges, and opportunities within the field of IoT in logistics and SCM.

An important consideration for future research is the strategic segmentation of publication analysis over different time periods. This approach can significantly deepen our understanding of how trends in IoT applications within logistics and SCM have evolved. By dividing the literature into distinct time segments—such as 5-year or 7-year intervals—researchers can trace the developmental trajectory of key themes and technologies more clearly. To effectively analyze these time-segmented data, scholars might employ strategic maps for each period. Conceptually, strategic mapping involves creating visual representations of the data that highlight the relationships between main topics, technologies, and methodologies over time [142]. These maps can visually trace the progression and shifts in focus within the field, illustrating how earlier studies have laid the groundwork for subsequent advancements and how new themes emerge and gain traction over time. Furthermore, dynamic topic modeling can be used as advanced technique to enhance the temporal segmentation of IoT research in logistics and SCM. This method allows researchers to analyze changes in topics over time within a corpus of documents. Consequently, future studies can apply dynamic topic modeling to quantitatively measure the shifts in discourse and focus, offering a more systematic and data-driven understanding of trends. Finally, future researchers have the opportunity to segment citation network analysis by different time periods [50]. This approach can offer insights into the most influential works and how their impact spreads throughout the academic community over time. Overall, by integrating these methodologies, future research can provide a comprehensive and time-sensitive analysis that maps out the historical and current landscape and predicts future directions and innovations in the field of IoT within logistics and SCM.

Data availability statement

Data is available upon request.

CRediT authorship contribution statement

Imen Zrelli: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Abderahman Rejeb:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was funded by the University of Jeddah, Jeddah, Saudi Arabia, under grant No. (UJ-23-DR-104). Therefore, the authors thank the University of Jeddah for its technical and financial support

References

[1] M. Ben-Daya, E. Hassini, Z. Bahroun, Internet of things and supply chain management: a literature review, Int. J. Prod. Res. 57 (2019) 4719–4742, https://doi.org/10.1080/00207543.2017.1402140.
[2] N. Kshetri, Can blockchain strengthen the internet of things? IT Prof 19 (2017) 68–72, https://doi.org/10.1109/MITP.2017.3051335.

[3] A. Rejeb, J.G. Keogh, H. Treiblmaier, Leveraging the internet of things and blockchain technology in supply chain management, Future Internet 11 (2019), https://doi.org/10.3390/fi11070161.
[4] L. Atzori, A. Iera, G. Morabito, The internet of things: a survey, Comput. Netw. 54 (2010) 2787–2805, https://doi.org/10.1016/j.comnet.2010.05.010.
[5] H.A. Abdulghani, N.A. Nijdam, D. Konstantas, Analysis on security and privacy guidelines: RFID-based IoT applications, IEEE Access 10 (2022) 131528–131554, https://doi.org/10.1109/ACCESS.2022.3227449.
[6] S. Evdokimov, B. Fabian, O. Günther, L. Ivantysynova, H. Ziekow, RFID and the Internet of Things: technology, applications, and security challenges, Found. Trends Technol. Inf. Oper. Manag. 4 (2011) 105–185, https://doi.org/10.1561/0200000020.
[7] M.M. Billah, S.S. Alam, M. Masukujjaman, M.H. Ali, Z.K.M. Makhbul, M.F.M. Salleh, Effects of Internet of Things, supply chain collaboration and ethical sensitivity on sustainable performance: moderating effect of supply chain dynamism, J. Enterp. Inf. Manag. 36 (2023) 1270–1295, https://doi.org/10.1108/JEIM-06-2022-0213.
[8] S. Cho, J. Kim, Smart logistics model on internet of things environment, Adv. Sci. Lett. 23 (2017) 1599–1602, https://doi.org/10.1166/asl.2017.8604.
[9] A. Rejeb, S. Simske, K. Rejeb, H. Treiblmaier, S. Zailani, Internet of Things research in supply chain management and logistics: a bibliometric analysis, Internet Things Neth 12 (2020), https://doi.org/10.1016/j.iot.2020.100318.
[10] S. Ahmed, T. Kalsoom, N. Ramzan, Z. Pervez, M. Azmat, B. Zeb, M. Ur Rehman, Towards supply chain visibility using internet of things: a dyadic analysis review, Sensors 21 (2021), https://doi.org/10.3390/s21124158.
[11] A. Ali, L. Feng, A.K. Bashir, S. El-Sappagh, S.H. Ahmed, M. Iqbal, G. Raja, Quality of service provisioning for heterogeneous services in cognitive radio-enabled internet of things, IEEE Trans. Netw. Sci. Eng 7 (2020) 328–342, https://doi.org/10.1109/TNSE.2018.2877646.
[12] M.J.M. Chowdhury, M.S. Ferdous, K. Biswas, N. Chowdhury, V. Muthukkumarasamy, A survey on blockchain-based platforms for IoT use-cases, Knowl. Eng. Rev. (2020), https://doi.org/10.1017/S0269888920000284.
[13] W. Kuandee, P. Nilsook, P. Wannapiroon, Asset supply chain management system-based IoT technology for higher education institutions, Int. J. Online Biomed. Eng. 15 (2019) 4–20, https://doi.org/10.3991/ijoe.v15i03.8533.
[14] H. Hellani, L. Sliman, A.E. Samhat, E. Exposito, Overview on the blockchain-based supply chain systematics and their scalability tools, Emerg. Sci. J 4 (2020) 45–69, https://doi.org/10.28991/esj-2021-SP1-04.
[15] H. Tran-Dang, D.-S. Kim, The physical internet in the era of digital transformation: perspectives and open issues, IEEE Access 9 (2021) 164613–164631, https://doi.org/10.1109/ACCESS.2021.3131562.
[16] H.A. Alsattar, N. Mourad, A. Zaidan, M. Deveci, S. Qahtan, V. Jayaraman, Z. Khalid, Developing IoT sustainable real-time monitoring devices for food supply chain systems based on climate change using circular intuitionistic fuzzy set, IEEE Internet Things J. (2023), https://doi.org/10.1109/JIOT.2023.3305910, 1–1.
[17] J. Chen, L. Zhang, Application of internet of things technology in logistics management systemof automobile supply chain, IPPTA Q. J. Indian Pulp Pap. Tech. - A 30 (2018) 49–56.
[18] A.W. Al-Khatib, Internet of things, big data analytics and operational performance: the mediating effect of supply chain visibility, J. Manuf. Technol. Manag. 34 (2023) 1–24, https://doi.org/10.1108/JMTM-08-2022-0310.
[19] X. Chen, C. He, Y. Chen, Z. Xie, Internet of Things (IoT)—blockchain-enabled pharmaceutical supply chain resilience in the post-pandemic era, Front. Eng. Manag. 10 (2023) 82–95, https://doi.org/10.1007/s42524-022-0233-1.
[20] V. Desingh, R. Baskaran, Internet of Things adoption barriers in the Indian healthcare supply chain: an ISM-fuzzy MICMAC approach, Int. J. Health Plann. Manag. 37 (2022) 318–351, https://doi.org/10.1002/hpm.3331.
[21] P. Agrawal, R. Narain, Analysis of enablers for the digitalization of supply chain using an interpretive structural modelling approach, Int. J. Prod. Perform. Manag. 72 (2023) 410–439, https://doi.org/10.1108/IJPPM-09-2020-0481.
[22] L. Haghnegahdar, S.S. Joshi, N.B. Dahotre, From IoT-based cloud manufacturing approach to intelligent additive manufacturing: industrial Internet of Things—an overview, Int. J. Adv. Manuf. Technol. 119 (2022) 1461–1478, https://doi.org/10.1007/s00170-021-08436-x.
[23] Y. Bendavid, L. Cassivi, Bridging the gap between RFID/EPC concepts, technological requirements and supply chain e-business processes, J. Theor. Appl. Electron. Commer. Res. 5 (2010) 1–16, https://doi.org/10.4067/S0718-18762010000300002.
[24] Y. Ernawan, C. Wijaya, F. Indriati, Application of internet of things to ensure improved performance within the internal supply chain unit of an organization, Int. J. Commun. Netw. Inf. Secur. 14 (2022) 121–135, https://doi.org/10.17762/IJCNIS.V14I1S.5620.
[25] M.M. Monsreal, R.B. Carmona-Benítez, Impact of IoT on supply chain performance, J. Appl. Res. Technol. 20 (2022) 584–593, https://doi.org/10.22201/icat.24486736e.2022.20.5.1957.
[26] Mordor Intelligence, Industrial internet of things market size & share analysis - growth trends & forecasts (2024 - 2029). https://www.mordorintelligence.com/industry-reports/industrial-internet-of-things-iiot-market, 2024. (Accessed 11 March 2024).
[27] Market Data Forecast, Industrial IoT Market Size, Share, Growth Forecast To 2029, Mark, Data Forecast, 2024. http://www.marketdataforecast.com/. (Accessed 11 March 2024).
[28] A. Rejeb, S. Simske, K. Rejeb, H. Treiblmaier, S. Zailani, Internet of Things research in supply chain management and logistics: a bibliometric analysis, Internet Things Neth 12 (2020), https://doi.org/10.1016/j.iot.2020.100318.
[29] M. Portugal Ferreira, A bibliometric study on Ghoshal's Managing across Borders, Multinatl. Bus. Rev. 19 (2011) 357–375, https://doi.org/10.1108/15253831111190180.
[30] N. Wang, H. Liang, Y. Jia, S. Ge, Y. Xue, Z. Wang, Cloud computing research in the IS discipline: a citation/co-citation analysis, Decis. Support Syst. 86 (2016) 35–47, https://doi.org/10.1016/j.dss.2016.03.006.
[31] N. Rezaee, S.M. Zanjirchi, N. Jalilian, S.M.H. Bamakan, Internet of things empowering operations management; A systematic review based on bibliometric and content analysis, Telemat. Inform. Rep. 11 (2023) 100096, https://doi.org/10.1016/j.teler.2023.100096.
[32] R. Katoch, IoT research in supply chain management and logistics: a bibliometric analysis using vosviewer software, Mater. Today Proc. 56 (2022) 2505–2515, https://doi.org/10.1016/j.matpr.2021.08.272.
[33] A. Aryal, Y. Liao, P. Nattuthurai, B. Li, The emerging big data analytics and IoT in supply chain management: a systematic review, Supply Chain Manag. Int. J. 25 (2018) 141–156, https://doi.org/10.1108/SCM-03-2018-0149.
[34] M. Ben-Daya, E. Hassini, Z. Bahroun, B.H. Banimfreg, The role of internet of things in food supply chain quality management: a review, Qual. Manag. J. 28 (2020) 17–40, https://doi.org/10.1080/10686967.2020.1838978.
[35] A. Rejeb, K. Rejeb, S. Zailani, H. Treiblmaier, K.J. Hand, Integrating the Internet of Things in the halal food supply chain: a systematic literature review and research agenda, Internet Things Neth 13 (2021), https://doi.org/10.1016/j.iot.2021.100361.
[36] D. Kumar, R. Kr Singh, R. Mishra, S. Fosso Wamba, Applications of the internet of things for optimizing warehousing and logistics operations: a systematic literature review and future research directions, Comput. Ind. Eng. 171 (2022) 108455, https://doi.org/10.1016/j.cie.2022.108455.
[37] S. Taj, A.S. Imran, Z. Kastrati, S.M. Daudpota, R.A. Memon, J. Ahmed, IoT-based supply chain management: a systematic literature review, Internet Things 24 (2023) 100982, https://doi.org/10.1016/j.iot.2023.100982.
[38] R.M.L. Rebelo, S.C.F. Pereira, M.M. Queiroz, The interplay between the Internet of things and supply chain management: challenges and opportunities based on a systematic literature review, Benchmarking Int. J. 29 (2021) 683–711, https://doi.org/10.1108/BIJ-02-2021-0085.
[39] K. Chapman, A.E. Ellinger, An evaluation of web of science, Scopus and google scholar citations in operations management, Int. J. Logist. Manag. 30 (2019) 1039–1053, https://doi.org/10.1108/IJLM-04-2019-0110.
[40] C. Villa-Collar, C. Álvarez-Peregrina, F. Hidalgo Santa Cruz, F.J. Povedano-Montero, Bibliometric study of scientific research on overnight orthokeratology, Eye Contact Lens 44 (2018) 344, https://doi.org/10.1097/ICL.0000000000000545.
[41] S.N. Wiyono, Y. Deliana, E. Wulandari, N.H. Kamarulzaman, The embodiment of muslim intention elements in buying halal food products: a literature review, Sustainability 14 (2022) 13163, https://doi.org/10.3390/su142013163.

[42] A. Rejeb, K. Rejeb, H. Treiblmaier, A. Appolloni, S. Alghamdi, Y. Alhasawi, M. Iranmanesh, The Internet of Things (IoT) in healthcare: taking stock and moving forward, Internet Things 22 (2023) 100721, https://doi.org/10.1016/j.iot.2023.100721.

[43] A. Rejeb, K. Rejeb, A. Abdollahi, S. Zailani, M. Iranmanesh, M. Ghobakhloo, Digitalization in food supply chains: a bibliometric review and key-route main path analysis, Sustain. Switz. 14 (2022), https://doi.org/10.3390/su14010083.

[44] R. Carè, O. Weber, What's in a name? Exploring the intellectual structure of social finance, Int. J. Emerg. Mark. (2023), https://doi.org/10.1108/IJOEM-07-2022-1142 ahead-of-print.

[45] D. Moher, A. Liberati, J. Tetzlaff, D.G. Altman, Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement, Ann. Intern. Med. 151 (2009) 264–269, https://doi.org/10.7326/0003-4819-151-4-200908180-00135.

[46] D. Denyer, D. Tranfield, Producing a systematic review, in: Sage Handb. Organ. Res. Methods, Sage Publications Ltd, Thousand Oaks, CA, 2009, pp. 671–689.

[47] A. Rejeb, K. Rejeb, A. Appolloni, H. Treiblmaier, M. Iranmanesh, Mapping the scholarly landscape of TikTok (Douyin): a bibliometric exploration of research topics and trends, Digit. Bus. 4 (2024) 100075, https://doi.org/10.1016/j.digbus.2024.100075.

[48] A. Abdollahi, K. Rejeb, A. Rejeb, M.M. Mostafa, S. Zailani, Wireless sensor networks in agriculture: insights from bibliometric analysis, Sustain. Switz. 13 (2021), https://doi.org/10.3390/su132112011.

[49] A. Rejeb, A. Abdollahi, K. Rejeb, H. Treiblmaier, Drones in agriculture: a review and bibliometric analysis, Comput. Electron. Agric. 198 (2022), https://doi.org/10.1016/j.compag.2022.107017.

[50] B. Fahimnia, J. Sarkis, H. Davarzani, Green supply chain management: a review and bibliometric analysis, Int. J. Prod. Econ. 162 (2015) 101–114, https://doi.org/10.1016/j.ijpe.2015.01.003.

[51] A. Rejeb, K. Rejeb, A. Appolloni, M. Iranmanesh, H. Treiblmaier, S. Jagtap, Exploring food supply chain trends in the COVID-19 era: a bibliometric review, Sustain. Switz. 14 (2022), https://doi.org/10.3390/su141912437.

[52] A. Rejeb, K. Rejeb, A. Abdollahi, F. Al-Turjman, H. Treiblmaier, The Interplay between the Internet of Things and agriculture: a bibliometric analysis and research agenda, Internet Things Neth 19 (2022), https://doi.org/10.1016/j.iot.2022.100580.

[53] C. Giatsidis, G. Nikolentzos, C. Zhang, J. Tang, M. Vazirgiannis, Rooted citation graphs density metrics for research papers influence evaluation, J. Informetr. 13 (2019) 757–768, https://doi.org/10.1016/j.joi.2019.03.006.

[54] A. Rejeb, K. Rejeb, K. Alnabulsi, S. Zailani, Tracing knowledge diffusion trajectories in scholarly bitcoin research: Co-word and main path analyses, J. Risk Financ. Manag. 16 (2023) 355, https://doi.org/10.3390/jrfm16080355.

[55] I. Soriano-Pinar, E. Díaz-Garrido, S. Bermejo-Olivas, Digital transformation for a circular economy: insights from co-word analysis, J. Technol. Transf. (2023), https://doi.org/10.1007/s10961-023-10041-x.

[56] P.-C. Lee, H.-N. Su, Investigating the structure of regional innovation system research through keyword co-occurrence and social network analysis, Innovation 12 (2010) 26–40, https://doi.org/10.5172/impp.12.1.26.

[57] H. Small, Co-citation in the scientific literature: a new measure of the relationship between two documents, J. Am. Soc. Inf. Sci. 24 (1973) 265–269, https://doi.org/10.1002/asi.4630240406.

[58] J.Y. Choi, S. Jeong, J.-K. Jung, Evolution of technology convergence networks in Korea: characteristics of temporal changes in R&D according to institution type, PLoS One 13 (2018) e0192195, https://doi.org/10.1371/journal.pone.0192195.

[59] Y. Cao, F. Jia, G. Manogaran, Efficient traceability systems of steel products using blockchain-based industrial internet of things, IEEE Trans. Ind. Inf. 16 (2020) 6004–6012, https://doi.org/10.1109/TII.2019.2942211.

[60] R. Jayaraman, K. Saleh, N. King, Improving opportunities in healthcare supply chain processes via the internet of things and blockchain technology, Int. J. Healthc. Inf. Syst. Inf. 14 (2019) 49–65, https://doi.org/10.4018/IJHISI.2019040104.

[61] M. Abdirad, K. Krishnan, Industry 4.0 in logistics and supply chain management: a systematic literature review, EMJ - Eng. Manag. J. (2020) 1–15, https://doi.org/10.1080/10429247.2020.1783935.

[62] D.R. Asokan, F.A. Huq, C.M. Smith, M. Stevenson, Socially responsible operations in the Industry 4.0 era: post-COVID-19 technology adoption and perspectives on future research, Int. J. Oper. Prod. Manag. 42 (2022) 185–217, https://doi.org/10.1108/IJOPM-01-2022-0069.

[63] C. Bai, H.A. Li, Y. Xiao, Industry 4.0 technologies: empirical impacts and decision framework, Prod. Oper. Manag. (2022), https://doi.org/10.1111/poms.13813.

[64] G. Alfian, M. Syafrudin, U. Farooq, M.R. Ma'arif, M.A. Syaekhoni, N.L. Fitriyani, J. Lee, J. Rhee, Improving efficiency of RFID-based traceability system for perishable food by utilizing IoT sensors and machine learning model, Food Control 110 (2020), https://doi.org/10.1016/j.foodcont.2019.107016.

[65] R. Alfred, J.H. Obit, C.P.-Y. Chin, H. Haviluddin, Y. Lim, Towards paddy rice smart farming: a review on big data, machine learning, and rice production tasks, IEEE Access 9 (2021) 50358–50380, https://doi.org/10.1109/ACCESS.2021.3069449.

[66] L. Chen, D.-G. Zhang, J. Zhang, T. Zhang, W.-J. Wang, Y.-H. Cao, A novel offloading approach of IoT user perception task based on quantum behavior particle swarm optimization, Future Gener. Comput. Syst. 141 (2023) 577–594, https://doi.org/10.1016/j.future.2022.12.016.

[67] S. Ding, A. Tukker, H. Ward, Opportunities and risks of internet of things (IoT) technologies for circular business models: a literature review, J. Environ. Manag. 336 (2023), https://doi.org/10.1016/j.jenvman.2023.117662.

[68] A. Rejeb, Z. Suhaiza, K. Rejeb, S. Seuring, H. Treiblmaier, The Internet of Things and the circular economy: a systematic literature review and research agenda, J. Clean. Prod. 350 (2022), https://doi.org/10.1016/j.jclepro.2022.131439.

[69] K. Addou, M.Y. El Ghoumari, S. Achkdir, M. Azzouazi, A decentralized model to ensure traceability and sustainability of the food supply chain by combining blockchain, IoT, and machine learning, Math. Model. Comput. 10 (2023) 498–510, https://doi.org/10.23939/mmc2023.02.498.

[70] T. Bosona, G. Gebresenbet, The role of blockchain technology in promoting traceability systems in agri-food production and supply chains, Sensors 23 (2023), https://doi.org/10.3390/s23115342.

[71] A.M. Aamer, M.A. Al-Awlaqi, I. Affia, S. Arumsari, N. Mandahawi, The internet of things in the food supply chain: adoption challenges, Benchmarking 28 (2021) 2521–2541, https://doi.org/10.1108/BIJ-07-2020-0371.

[72] H. Ahmad Tarmizi, N.H. Kamarulzaman, A. Abd Rahman, R. Atan, Adoption of internet of things among Malaysian halal agro-food smes and its challenges, Food Res. 4 (2020) 256–265, https://doi.org/10.26656/fr.2017.4(S1).S26.

[73] I. Lee, K. Lee, The internet of things (IoT): applications, investments, and challenges for enterprises, bus, Horiz. 58 (2015) 431–440, https://doi.org/10.1016/j.bushor.2015.03.008.

[74] S. Wolfert, L. Ge, C. Verdouw, M.-J. Bogaardt, Big data in smart farming – a review, Agric. Syst. 153 (2017) 69–80, https://doi.org/10.1016/j.agsy.2017.01.023.

[75] A.G. Frank, L.S. Dalenogare, N.F. Ayala, Industry 4.0 technologies: implementation patterns in manufacturing companies, Int. J. Prod. Econ. 210 (2019) 15–26, https://doi.org/10.1016/j.ijpe.2019.01.004.

[76] E. Hofmann, M. Rüsch, Industry 4.0 and the current status as well as future prospects on logistics, Comput. Ind. 89 (2017) 23–34, https://doi.org/10.1016/j.compind.2017.04.002.

[77] R.H. Weber, Internet of Things - new security and privacy challenges, Comput. Law Secur. Rev. 26 (2010) 23–30, https://doi.org/10.1016/j.clsr.2009.11.008.

[78] A. Kusiak, Smart manufacturing, Int. J. Prod. Res. 56 (2018) 508–517, https://doi.org/10.1080/00207543.2017.1351644.

[79] K.-D. Thoben, S.A. Wiesner, T. Wuest, "Industrie 4.0" and smart manufacturing-a review of research issues and application examples, Int. J. Autom. Technol. 11 (2017) 4–16, https://doi.org/10.20965/ijat.2017.p0004.

[80] A.B. Lopes de Sousa Jabbour, C.J.C. Jabbour, M. Godinho Filho, D. Roubaud, Industry 4.0 and the circular economy: a proposed research agenda and original roadmap for sustainable operations, Ann. Oper. Res. 270 (2018) 273–286, https://doi.org/10.1007/s10479-018-2772-8.

[81] Y. Feng, Q. Zhu, K.-H. Lai, Corporate social responsibility for supply chain management: a literature review and bibliometric analysis, J. Clean. Prod. 158 (2017) 296–307, https://doi.org/10.1016/j.jclepro.2017.05.018.

[82] K. Börner, C. Chen, K.W. Boyack, Visualizing knowledge domains, Annu. Rev. Inf. Sci. Technol. 37 (2003) 179–255, https://doi.org/10.1002/aris.1440370106.

[83] J. Sander, M. Ester, H.-P. Kriegel, X. Xu, Density-based clustering in spatial databases: the algorithm GDBSCAN and its applications, Data Min. Knowl. Discov. 2 (1998) 169–194, https://doi.org/10.1023/A:1009745219419.
[84] A. Alwadi, A. Gawanmeh, S. Parvin, J.N. Al-Karaki, Smart solutions for RFID based inventory management systems: a survey, Scalable Comput 18 (2017) 347–360, https://doi.org/10.12694/scpe.v18i4.1333.
[85] T. Fan, F. Tao, S. Deng, S. Li, Impact of RFID technology on supply chain decisions with inventory inaccuracies, Int. J. Prod. Econ. 159 (2015) 117–125, https://doi.org/10.1016/j.ijpe.2014.10.004.
[86] C. Li, E. Tanghe, P. Suanet, D. Plets, J. Hoebeke, E. De Poorter, W. Joseph, ReLoc 2.0: UHF-RFID relative localization for drone-based inventory management, IEEE Trans. Instrum. Meas. 70 (2021), https://doi.org/10.1109/TIM.2021.3069377.
[87] S.-L. Chen, Y.-Y. Chen, C. Hsu, A new approach to integrate internet-of-things and software-as-a-service model for logistic systems: a case study, Sens. Switz. 14 (2014) 6144–6164, https://doi.org/10.3390/s140406144.
[88] R. Davis, M. Vochozka, J. Vrbka, O. Neguriță, Industrial artificial intelligence, smart connected sensors, and big data-driven decision-making processes in internet of things-based real-time production logistics, Econ. Manag. Financ. Mark. 15 (2020) 9–15, https://doi.org/10.22381/EMFM15320201.
[89] F. Goodarzian, A. Navaei, B. Ehsani, P. Ghasemi, J. Muñuzuri, Designing an integrated responsive-green-cold vaccine supply chain network using Internet-of-Things: artificial intelligence-based solutions, Ann. Oper. Res. 328 (2023) 531–575, https://doi.org/10.1007/s10479-022-04713-4.
[90] T. Ahmed, C.L. Karmaker, S.B. Nasir, M.A. Moktadir, S.K. Paul, Modeling the artificial intelligence-based imperatives of industry 5.0 towards resilient supply chains: a post-COVID-19 pandemic perspective, Comput. Ind. Eng. 177 (2023), https://doi.org/10.1016/j.cie.2023.109055.
[91] D. Gupta, S. Bhatt, M. Gupta, A.S. Tosun, Future smart connected communities to fight COVID-19 outbreak, Internet Things Neth 13 (2021), https://doi.org/10.1016/j.iot.2020.100342.
[92] B. Mohapatra, S. Tripathy, D. Singhal, R. Saha, Significance of digital technology in manufacturing sectors: examination of key factors during Covid-19, Res. Transp. Econ. 93 (2022), https://doi.org/10.1016/j.retrec.2021.101134.
[93] A.Z. Abideen, V.P.K. Sundram, J. Pyeman, A.K. Othman, S. Sorooshian, Digital twin integrated reinforced learning in supply chain and logistics, Logistics 5 (2021), https://doi.org/10.3390/logistics5040084.
[94] I. Abouzid, R. Saidi, Digital twin implementation approach in supply chain processes, Sci. Afr. 21 (2023), https://doi.org/10.1016/j.sciaf.2023.e01821.
[95] S. Abdallah, N. Nizamuddin, Blockchain-based solution for pharma supply chain industry, Comput. Ind. Eng. 177 (2023), https://doi.org/10.1016/j.cie.2023.108997.
[96] B. Esmaeilian, J. Sarkis, K. Lewis, S. Behdad, Blockchain for the future of sustainable supply chain management in Industry 4.0, Resour. Conserv. Recycl. 163 (2020), https://doi.org/10.1016/j.resconrec.2020.105064.
[97] M. Ahmed, C. Taconet, M. Ould, S. Chabridon, A. Bouzeghoub, IoT data qualification for a logistic chain traceability smart contract, Sensors 21 (2021), https://doi.org/10.3390/s21062239.
[98] M. Grida, N.A. Mostafa, Are smart contracts too smart for Supply Chain 4.0? A blockchain framework to mitigate challenges, J. Manuf. Technol. Manag. 34 (2023) 644–665, https://doi.org/10.1108/JMTM-09-2021-0359.
[99] E.M. Adere, Blockchain in healthcare and IoT: a systematic literature review, Array 14 (2022), https://doi.org/10.1016/j.array.2022.100139.
[100] S.H. Awan, S. Ahmad, Y. Khan, N. Safwan, S.S. Qurashi, M.Z. Hashim, A combo smart model of blockchain with the internet of things (IoT) for the transformation of agriculture sector, Wirel. Pers. Commun. 121 (2021) 2233–2249, https://doi.org/10.1007/s11277-021-08820-6.
[101] N. Ada, Y. Kazancoglu, M.D. Sezer, C. Ede-Senturk, I. Ozer, M. Ram, Analyzing barriers of circular food supply chains and proposing industry 4.0 solutions, Sustain. Switz. 13 (2021), https://doi.org/10.3390/su13126812.
[102] S. Chen, B. Li, Z. Chen, Y. Zhang, C. Wang, C. Tao, Novel strong-PUF-based authentication protocols leveraging shamir's secret sharing, IEEE Internet Things J. 9 (2022) 14408–14425, https://doi.org/10.1109/JIOT.2021.3065836.
[103] M. Alalhareth, S.-C. Hong, An improved mutual information feature selection technique for intrusion detection systems in the internet of medical things, Sensors 23 (2023), https://doi.org/10.3390/s23104971.
[104] S. Al-E'mari, M. Anbar, Y. Sanjalawe, S. Manickam, I. Hasbullah, Intrusion detection systems using blockchain technology: a review, issues and challenges, Comput. Syst. Sci. Eng. 40 (2022) 87–112, https://doi.org/10.32604/CSSE.2022.017941.
[105] R. Sudarmani, K. Venusamy, S. Sivaraman, P. Jayaraman, K. Suriyan, M. Alagarsamy, Machine to machine communication enabled internet of things: a review, Int. J. Reconfigurable Embed. Syst. 11 (2022) 126–134, https://doi.org/10.11591/ijres.v11.i2.pp126-134.
[106] R. Aknin, Y. Bentaleb, Enhanced MQTT architecture for smart supply chain, Int. J. Adv. Comput. Sci. Appl. 14 (2023) 861–869, https://doi.org/10.14569/IJACSA.2023.0140495.
[107] S. Alyahya, W.U. Khan, S. Ahmed, S.N.K. Marwat, S. Habib, Cyber secure framework for smart agriculture: robust and tamper-resistant authentication scheme for IoT devices, electron, Switz. 11 (2022), https://doi.org/10.3390/electronics11060963.
[108] R.Y. Zhong, C. Xu, C. Chen, G.Q. Huang, Big data analytics for physical internet-based intelligent manufacturing shop floors, Int. J. Prod. Res. 55 (2017) 2610–2621, https://doi.org/10.1080/00207543.2015.1086037.
[109] J. Hopkins, P. Hawking, Big data analytics and IoT in logistics: a case study, Int. J. Logist. Manag. 29 (2018) 575–591, https://doi.org/10.1108/IJLM-05-2017-0109.
[110] R.Y. Zhong, S. Lan, C. Xu, Q. Dai, G.Q. Huang, Visualization of RFID-enabled shopfloor logistics big data in cloud manufacturing, Int. J. Adv. Manuf. Technol. 84 (2016) 5–16, https://doi.org/10.1007/s00170-015-7702-1.
[111] D. Zhu, IOT and big data based cooperative logistical delivery scheduling method and cloud robot system, Future Gener. Comput. Syst. 86 (2018) 709–715, https://doi.org/10.1016/j.future.2018.04.081.
[112] L. He, M. Xue, B. Gu, Internet-of-things enabled supply chain planning and coordination with big data services: certain theoretic implications, J. Manag. Sci. Eng. 5 (2020) 1–22, https://doi.org/10.1016/j.jmse.2020.03.002.
[113] C.N. Verdouw, J. Wolfert, A.J.M. Beulens, A. Rialland, Virtualization of food supply chains with the internet of things, J. Food Eng. 176 (2016) 128–136, https://doi.org/10.1016/j.jfoodeng.2015.11.009.
[114] Y. Zhang, L. Zhao, C. Qian, Modeling of an IoT-enabled supply chain for perishable food with two-echelon supply hubs, Ind. Manag. Data Syst. 117 (2017) 1890–1905, https://doi.org/10.1108/IMDS-10-2016-0456.
[115] S.S. Kamble, A. Gunasekaran, H. Parekh, S. Joshi, Modeling the internet of things adoption barriers in food retail supply chains, J. Retail. Consum. Serv. 48 (2019) 154–168, https://doi.org/10.1016/j.jretconser.2019.02.020.
[116] Z. Pang, Q. Chen, W. Han, L. Zheng, Value-centric design of the internet-of-things solution for food supply chain: value creation, sensor portfolio and information fusion, Inf. Syst. Front 17 (2015) 289–319, https://doi.org/10.1007/s10796-012-9374-9.
[117] Y. Bouzembrak, M. Klüche, A. Gavai, H.J.P. Marvin, Internet of Things in food safety: literature review and a bibliometric analysis, Trends Food Sci. Technol. 94 (2019) 54–64, https://doi.org/10.1016/j.tifs.2019.11.002.
[118] J. Wang, H. Yue, Food safety pre-warning system based on data mining for a sustainable food supply chain, Food Control 73 (2017) 223–229, https://doi.org/10.1016/j.foodcont.2016.09.048.
[119] Z. Li, G. Liu, L. Liu, X. Lai, G. Xu, IoT-based tracking and tracing platform for prepackaged food supply chain, Ind. Manag. Data Syst. 117 (2017) 1906–1916, https://doi.org/10.1108/IMDS-11-2016-0489.
[120] S.-Y. Lin, L. Zhang, J. Li, L.-L. Ji, Y. Sun, A survey of application research based on blockchain smart contract, Wirel. Netw. 28 (2022) 635–690, https://doi.org/10.1007/s11276-021-02874-x.
[121] T.M. Hewa, Y. Hu, M. Liyanage, S.S. Kanhare, M. Ylianttila, Survey on blockchain-based smart contracts: technical aspects and future research, IEEE Access 9 (2021) 87643–87662, https://doi.org/10.1109/ACCESS.2021.3068178.
[122] H. Hasan, E. AlHadhrami, A. AlDhaheri, K. Salah, R. Jayaraman, Smart contract-based approach for efficient shipment management, Comput. Ind. Eng. 136 (2019) 149–159, https://doi.org/10.1016/j.cie.2019.07.022.

[123] T.H. Pranto, A.A. Noman, A. Mahmud, A.B. Haque, Blockchain and smart contract for IoT enabled smart agriculture, PeerJ Comput. Sci. 7 (2021) 1–29, https://doi.org/10.7717/PEERJ-CS.407.
[124] A. Khatoon, A blockchain-based smart contract system for healthcare management, Electron. Switz 9 (2020), https://doi.org/10.3390/electronics9010094.
[125] A. Albizri, D. Appelbaum, Trust but verify: the oracle paradox of blockchain smart contracts, J. Inf. Syst. 35 (2021) 1–16, https://doi.org/10.2308/ISYS-19-024.
[126] B. Tjahjono, C. Esplugues, E. Ares, G. Pelaez, What does industry 4.0 mean to supply chain? Procedia Manuf. 13 (2017) 1175–1182, https://doi.org/10.1016/j.promfg.2017.09.191.
[127] E. Manavalan, K. Jayakrishna, A review of Internet of Things (IoT) embedded sustainable supply chain for industry 4.0 requirements, Comput. Ind. Eng. 127 (2019) 925–953, https://doi.org/10.1016/j.cie.2018.11.030.
[128] L. Barreto, A. Amaral, T. Pereira, Industry 4.0 implications in logistics: an overview, Procedia Manuf. 13 (2017) 1245–1252, https://doi.org/10.1016/j.promfg.2017.09.045.
[129] S. Winkelhaus, E.H. Grosse, Logistics 4.0: a systematic review towards a new logistics system, Int. J. Prod. Res. 58 (2020) 18–43, https://doi.org/10.1080/00207543.2019.1612964.
[130] H. Fatorachian, H. Kazemi, Impact of Industry 4.0 on supply chain performance, Prod. Plan. Control 32 (2021) 63–81, https://doi.org/10.1080/09537287.2020.1712487.
[131] I.H. Sarker, Y.B. Abushark, F. Alsolami, A.I. Khan, IntruDTree: a machine learning based cyber security intrusion detection model, Symmetry 12 (2020), https://doi.org/10.3390/SYM12050754.
[132] A. Abbas, M.A. Khan, S. Latif, M. Ajaz, A.A. Shah, J. Ahmad, A new ensemble-based intrusion detection system for internet of things, Arab. J. Sci. Eng. 47 (2022) 1805–1819, https://doi.org/10.1007/s13369-021-06086-5.
[133] Z.E. Huma, S. Latif, J. Ahmad, Z. Idrees, A. Ibrar, Z. Zou, F. Alqahtani, F. Baothman, A hybrid deep random neural network for cyberattack detection in the industrial internet of things, IEEE Access 9 (2021) 55595–55605, https://doi.org/10.1109/ACCESS.2021.3071766.
[134] P.K.R. Maddikunta, Q.-V. Pham, P. B, N. Deepa, K. Dev, T.R. Gadekallu, R. Ruby, M. Liyanage, Industry 5.0: a survey on enabling technologies and potential applications, J. Ind. Inf. Integr. 26 (2022), https://doi.org/10.1016/j.jii.2021.100257.
[135] S.S. Kamble, A. Gunasekaran, H. Parekh, V. Mani, A. Belhadi, R. Sharma, Digital twin for sustainable manufacturing supply chains: current trends, future perspectives, and an implementation framework, Technol. Forecast. Soc. Change 176 (2022), https://doi.org/10.1016/j.techfore.2021.121448.
[136] M. Andronie, G. Lăzăroiu, R. Ștefănescu, C. Uță, I. Dijmărescu, Sustainable, smart, and sensing technologies for cyber-physical manufacturing systems: a systematic literature review, Sustain. Switz. 13 (2021), https://doi.org/10.3390/su13105495.
[137] T.D. Moshood, G. Nawanir, S. Sorooshian, O. Okfalisa, Digital twins driven supply chain visibility within logistics: a new paradigm for future logistics, Appl. Syst. Innov. 4 (2021), https://doi.org/10.3390/asi4020029.
[138] E.L. Alvarez-Aros, C.A. Bernal-Torres, Technological competitiveness and emerging technologies in industry 4.0 and industry 5.0, An. Acad. Bras, Cienc. 93 (2021), https://doi.org/10.1590/0001-3765202120191290.
[139] A. Verma, P. Bhattacharya, N. Madhani, C. Trivedi, B. Bhushan, S. Tanwar, G. Sharma, P.N. Bokoro, R. Sharma, Blockchain for industry 5.0: vision, opportunities, key enablers, and future directions, IEEE Access 10 (2022) 69160–69199, https://doi.org/10.1109/ACCESS.2022.3186892.
[140] M.K. Linnenluecke, M. Marrone, A.K. Singh, Conducting systematic literature reviews and bibliometric analyses, Aust. J. Manag. 45 (2020) 175–194, https://doi.org/10.1177/0312896219877678.
[141] C. Chen, L. Leydesdorff, Patterns of connections and movements in dual-map overlays: a new method of publication portfolio analysis, J. Assoc. Inf. Sci. Technol. 65 (2014) 334–351, https://doi.org/10.1002/asi.22968.
[142] A. Rejeb, K. Rejeb, I. Zrelli, E. Süle, M. Iranmanesh, Blockchain technology in the renewable energy sector: a co-word analysis of academic discourse, Heliyon 10 (8) (2024) e29600. https://www.cell.com/heliyon/fulltext/S2405-8440(24)05631-7. (Accessed 7 May 2024), 1-25.